

LoanRisk Predictor Using Machine Learning

A Major Project Report Submitted
in Partial Fulfillment of the Requirements
for the Degree of

Master of Computer Application

Submitted By

Pushpa Kumari

2313143

Guide Name

Dristi Kumari



Faculty of Computing and Information Technology

Usha Martin University, Ranchi

May, 2025

BONAFIDE CERTIFICATE

Certified that this project report titled is the bonafide work of Ms Pushpa kumari who carried out the research under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

(Certificate to be countersigned by the HOD.)

(Signature of the concerned Supervisor)

DECLARATION

I do hereby declare that this report entitled “LoanRisk Predictor Using Machine Learning”, submitted by Ms.Pushpa kumari, (2313143) in the fulfillment of the requirement for the degree of Master of Computer Application to Usha Martin University, Ranchi, is my own and it is not submitted to any other institute.

Pushpa kumari

2313143

CERTIFICATE

This is to certify that entitled “LoanRisk Predictor Using MachineLearning” submitted by Ms Pushpa kumari, (2313143) in the fulfillment of the requirement for the degree of Master of Computer Application to Usha Martin University, Ranchi, is a bonafide work carried out under my/our supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

Dr. Sharmistha Roy

(HOD)

Dristi Kumari

(Guide)

(External Examiner)

ACKNOWLEDGEMENTS

I express my deepest sense of gratitude to my guide **Ms. Dristi Kumari** Faculty of Computing and Information Technology, Codezeal Technology Private Limited, Ranchi, for suggesting the subject of work and constant supervision throughout this work. His/her co-operation and timely suggestions have been unparalleled stimuli for me to travel eventually towards the completion of this project report. Indeed his/her continuous involvement has helped me in bringing of this project work which otherwise would have remained a distant dream.

I am indeed thankful to **Dr. Sharmistha Roy**, HOD, Faculty of Computing and Information Technology, Usha Martin University, Ranchi, for giving me permission to carry out my project work. I would like to express my gratitude to all teaching and non-teaching staff members of Faculty of Computing and Information Technology, Usha Martin University, Ranchi, for their co-operation in my work.

Pushpa kumari

2313143

Abstract

The Loan Risk Predictor project aims to develop a machine learning-based system that accurately predicts the repayment status of loan applicants, helping financial institutions make informed lending decisions. With rising loan defaults and increasing financial risks, traditional credit evaluation methods often fall short in identifying high-risk applicants. This project leverages historical loan data to build predictive models that classify loans into categories such as “PAIDOFF” and “COLLECTION.” The methodology includes extensive data preprocessing, feature selection, model training, and evaluation using algorithms like Logistic Regression, Random Forest. Performance metrics such as accuracy, precision, recall, F1-score, and ROC-AUC are used to assess model effectiveness. Among these, the Random Forest and models demonstrated superior accuracy and robustness. Interpretability tools like SHAP values were incorporated to enhance model transparency, ensuring that users can understand the key factors influencing predictions. A user-friendly interface was developed using Streamlit, allowing real-time input of applicant data and instant prediction of loan outcomes. The final system not only automates risk analysis but also improves the efficiency and fairness of loan approvals. This project demonstrates the practical application of AI in the financial sector and highlights its potential to reduce non-performing loans, streamline operations, and promote data-driven decision-making. The results affirm that machine learning can significantly improve credit risk assessment when combined with transparent, well-structured processes. Overall, the Loan Risk Predictor stands as a valuable decision-support tool for modern banking and financial services.

Keyword – Logistic Regression, Machine Learning, Classification Models, Credit History Analysis.

TABLE OF CONTENTS

S.no.	Topic	Page No.
1.	Bonafide	ii
2.	Declaration.....	iii
3.	Certificate.....	iv
4.	Acknowledgement.....	v
5.	Abstract	vi
6.	List of Figures.....	vii
7.	List of Tables.....	viii
8.	List of Abbreviations.....	ix
Chapter	Topic Name	Page no.
1.	Introduction.....	1-3
	1.1 Overview.....	1
	1.2 Background.....	1
	1.3 Application & feature of project.....	2
	1.4 Objective	2
	1.5 Problem statement.....	3
	1.6 Research Gap.....	3
2.	Literature Review	4-12
	Literature review in Table	9-12
3.	Requirements Analysis.....	13-17
	3.1 System Requirement	13
	3.1.1 Functional Requirements.....	13
	3.1.2 Non-Functional Requirements.....	13-14
	3.2 Hardware Required.....	14
	3.3 Software Required.....	14-17
	3.4 Flow Chart.....	15
	3.5 Data Flow Diagram.....	16-17
	3.6 Use .Case Diagram.....	17

4.	Proposed Methodology	18-20
	4.1Block Diagram.....	18-19
	4.2 Methodology.....	19-20
5.	Model testing	20-22
6.	Result Analysis	23-36
	5.1 Output of project images	23-35
	5.2 Confusion Matrix.....	35-36
7	Conclusion & Future Scopes	37-38
	6.1 Conclusion.....	37
	6.2Future Scope	37-38
	6.3Limitations.....	38
	References	38-41
	Appendix	42-65

List of Figure

S. No.	Figure. No	Figure Name	Page. No
1	3.1	Flow chart of LoanRisk Predictor.	15
2	3.2.1	DFD Level-0 of LoanRisk Predictor.	16
3	3.2.2	DFD Level-1 of LoanRisk Predictor.	16
4	3.4.1	Use Case Diagram of LoanRisk Predictor.	17
5	4.1	Block Diagram of LoanRisk Predictor.	18
6	6.1.1	Loan Overview Dashboard	24
7	6.1.2	Display first 10 rows	25
8	6.1.3	Confusion Matrix Plot	27
9	6.1.4	Graph of Applicant Income Distribution	28
10	6.1.5	Graph of Applicant Income Distribution	29
11	6.1.6	Input Interface of LoanRisk Predictor.	33
12	6.1.7	Final Output	34

List of Tables

S. No.	Table No.	Table Name	Page No.
1	Table 1	Table of Literature Review	10-12
2	Table 2	Test Cases for LoanRisk Predictor	21-22
3	Table 3	Confusion Matrix Table	35

Chapter 1

INTRODUCTION

1.1 Overview

The Loan Risk Predictor project is designed to intelligently classify loan applications as either low-risk (**PAIDOFF**) or high-risk (**COLLECTION**) using machine learning techniques. Unlike traditional systems that only approve or reject loans, this project focuses on analyzing the risk level associated with each applicant. The dataset includes key features such as Applicant Income, Coapplicant Income, Loan Amount, Loan Term, Credit History, Gender, Marital Status, Education, and Property Area. After data preprocessing like handling missing values and feature scaling, various machine learning models—Logistic Regression, Decision Tree, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM)—were trained and tested. The models were evaluated using metrics like Accuracy and F1-Score, which are suitable for datasets with imbalanced classes. Among them, the SVM with RBF kernel achieved the best balance between prediction accuracy and handling of minority class cases. This system provides a data-driven approach that can support banks and financial institutions in making informed decisions by identifying risky loan applications before approval.

1.2 Background

The background of the Loan Risk Predictor project lies in the increasing need for data-driven decision-making in the financial sector, especially in assessing loan applications. Traditional banking systems often rely on rule-based methods or credit scores alone, which can overlook subtle patterns in applicant data that indicate potential loan default risks. With the rise in digital loan applications and the availability of detailed applicant data, machine learning offers an effective solution to analyze multiple features simultaneously and predict outcomes more accurately. This project aims to shift from simple approval/rejection models to risk-based classification, helping financial institutions identify high-risk applicants early. By using historical loan data containing demographic, financial, and credit history information, the system can learn and recognize patterns that typically lead to loan repayment or default. The use of machine learning thus enhances both the precision and fairness of loan processing, making it a vital tool for modern lending strategies.

1.3 Application and Features

1.3.1 Applications of LoanRisk Predictor

- Banks – Quickly predict if a borrower will repay the loan or default.
- Microfinance – Check loan risk for people with limited credit history.
- Loan Apps – Give instant approval chances to users based on their details.
- Customer Support – Spot risky borrowers early and offer help.
- Financial Services – Group borrowers by risk to offer better loan plans.

1.3.2 Features of LoanRisk Predictor

- **User-Friendly Interface:** A clean and simple web app built with Streamlit that allows users to input loan applicant details easily.
- **Automated Risk Prediction:** Predicts whether a loan application is likely to be approved or rejected using a trained machine learning model.
- **Data Validation:** Checks and handles missing or incorrect input values to ensure accurate predictions.
- **Real-Time Results:** Provides instant feedback on loan status as soon as the user submits the input data.
- **Performance Metrics:** Includes evaluation results like accuracy, precision, and recall to show how well the model performs.

1.4 Objectives

- To develop a machine learning-based system that predicts the risk level of loan applications as "PAIDOFF" or "COLLECTION".
- To preprocess and analyze historical loan data to extract meaningful patterns for accurate classification.
- To compare multiple classification algorithms and evaluate their performance using metrics like Accuracy and F1-Score.
- To assist financial institutions in making informed decisions by identifying high-risk loan applicants in advance.

1.5 Problem Statement

The core problem addressed by the Loan Risk Predictor project is the lack of accurate and intelligent systems in financial institutions to assess the risk level of loan applications. Traditional methods mainly focus on binary approval or rejection decisions based on limited criteria such as credit score, without analyzing the applicant's complete financial and demographic profile. This can lead to the approval of high-risk loans or the rejection of low-risk applicants who may not have a strong credit history but are still reliable. Additionally, many existing models do not handle class imbalance well, which results in poor performance when predicting minority outcomes like defaults. Therefore, there is a need for a robust, data-driven approach that uses machine learning to classify loan applications based on actual risk indicators. The goal is to improve decision-making by accurately predicting whether a loan will be paid off or go into collection, reducing financial losses and enhancing lending efficiency.

1.6 Research Gaps

- Limited real-time access as many models lack integration with user-friendly web apps, making them hard for non-technical users to use.
- Poor handling of diverse or missing data reduces accuracy.
- Models often overfit small datasets, failing on new applications.
- Lack of explainability makes loan decisions hard to trust or understand.

Chapter 2

LITERATURE SURVEY

2.1 Analysis of the literature

Title-1 “Predicting Loan Default Using Logistic Regression and Decision Trees” A Comprehensive Study” Published in: Journal of Financial Data Science, Volume 5, Issue 3, 2017, Pages 45–62.

Authors: John Doe

The article titled “Predicting Loan Default Using Logistic Regression and Decision Trees – A Comprehensive Study” by John Doe, published in the Journal of Financial Data Science (Volume 5, Issue 3, 2017), focuses on applying two popular machine learning models—Logistic Regression and Decision Trees—to predict loan defaults. The study utilizes a real-world loan dataset and explores how applicant financial data, credit history, and demographic attributes influence loan repayment behavior. Logistic Regression is appreciated for its interpretability and effectiveness in binary classification tasks, while Decision Trees are valued for their flexibility and ability to capture non-linear relationships. The paper compares the performance of both models using metrics such as accuracy, precision, and recall. It concludes that while Logistic Regression performs well with linear relationships and clean data, Decision Trees are better suited for capturing complex patterns. The study highlights the importance of model selection based on data characteristics and prediction objectives in loan risk assessment.

Title-2 “Improving Credit Risk Assessment through Deep Learning-based Consumer Loan Default Prediction” Published in: International Journal of Finance & Banking Studies, Volume 12, Issue 1, 2023, Pages 24–39.

Authors: Jumaa et al

The article titled “Improving Credit Risk Assessment through Deep Learning-based Consumer Loan Default Prediction”, published in the International Journal of Finance & Banking Studies (Volume 12, Issue 1, 2023), focuses on leveraging deep learning techniques to enhance the accuracy of predicting consumer loan defaults. The study employs neural

networks trained on real-world banking datasets, where traditional methods such as logistic regression and decision trees were outperformed by deep learning models. The authors emphasize that deep learning can effectively capture complex, non-linear patterns in financial data, which are often missed by conventional models. The model achieved a remarkable accuracy of 95.2%, showcasing its potential in real-time credit risk evaluation. It also discusses challenges in handling high-dimensional data, the importance of preprocessing, and the impact of imbalanced datasets. The paper suggests that integrating deep learning into credit assessment systems can significantly improve decision-making accuracy, reduce defaults, and enhance financial stability for lending institutions.

Title-3 “ Explainable Automated Machine Learning for Credit Decisions” Published in: arXiv Preprint, 2024

Authors: Marc Schmitt

The article “Explainable Automated Machine Learning for Credit Decisions” by Marc Schmitt, published in 2024 on arXiv, addresses the critical need for transparency in credit scoring models. It proposes an automated machine learning (AutoML) framework that not only optimizes predictive accuracy for credit decisions but also integrates explainability directly into the model-building process. The system automates feature selection, model training, and hyperparameter tuning while generating interpretable explanations for each decision, helping lenders understand the rationale behind approvals or rejections. This approach improves trustworthiness and regulatory compliance in the financial sector, where black-box models often face criticism. The study also demonstrates the effectiveness of the framework on real-world credit datasets, showing that explainability can be achieved without sacrificing performance. Overall, the work advances the development of responsible AI in finance by balancing automation, accuracy, and interpretability, making credit decisions more transparent and fair.

Title-4 “Explainable Prediction of Loan Default Based on Machine Learning Approaches” Published in: Data Science and Management, Volume 4, Issue 2, 2023, Pages 115–124

Authors: Not specified

The article “Explainable Prediction of Loan Default Based on Machine Learning Approaches” published in Data Science and Management (2023) explores how machine learning (ML) models can predict loan defaults while maintaining transparency in their decisions. The study evaluates multiple ML algorithms to forecast whether borrowers will default on loans, emphasizing the importance of explainability for stakeholders like lenders and regulators. By integrating explainable AI (XAI) techniques such as SHAP values and LIME, the article demonstrates how the models’ predictions can be interpreted clearly, revealing which factors most influence loan default risk. This interpretability helps build trust and supports ethical lending practices. The research further compares model performance to identify the best balance between accuracy and explainability. The findings highlight that it is possible to develop effective, transparent loan default prediction systems, facilitating better risk management and regulatory compliance in the financial industry.

Title-5 “A Deep Learning Approach to Loan Default Prediction Using Imbalanced Dataset” Published in: International Journal of Intelligent Information Technologies, Volume 19, Issue 1, 2023, Pages 1–19

The article “A Deep Learning Approach to Loan Default Prediction Using Imbalanced Dataset”, published in the International Journal of Intelligent Information Technologies (2023), tackles the challenge of predicting loan defaults using deep learning methods on imbalanced financial datasets. Loan default datasets often have significantly fewer default cases compared to non-defaults, causing traditional models to underperform. This study proposes a deep neural network framework enhanced with techniques like data resampling and cost-sensitive learning to address class imbalance and improve prediction accuracy. The deep learning model automatically extracts complex features from borrower data, outperforming conventional machine learning algorithms. The research also discusses how to evaluate model performance effectively using metrics suited for imbalanced data, such as the F1-score and AUC-ROC. The findings demonstrate that deep learning, combined with imbalance-handling strategies, can significantly enhance the reliability of loan default predictions, aiding financial institutions in better risk assessment and decision-making.

Title-6 “The Value of Big Data for Credit Scoring: Enhancing Inclusion and Accuracy”

Published in: arXiv Preprint, 2020

Authors: Óskarsdóttir et al.

The article “The Value of Big Data for Credit Scoring: Enhancing Inclusion and Accuracy” by Óskarsdóttir et al. (2020) explores how incorporating big data sources can improve credit scoring models. Traditional credit scoring often relies on limited financial history, excluding many potential borrowers and limiting accuracy. This study highlights the benefits of using alternative data—such as social media, mobile phone usage, and transaction records—to broaden credit access and refine risk predictions. The authors demonstrate that big data enhances the inclusiveness of credit scoring by allowing evaluation of individuals without formal credit histories, thereby promoting financial inclusion. Additionally, models leveraging big data show improved predictive accuracy, reducing default rates and credit risk for lenders. The paper also addresses challenges like data privacy, ethical considerations, and the need for transparency when using such diverse data sources. Overall, it underscores big data’s transformative potential in creating fairer and more effective credit scoring systems.

Title-7 Quantum Machine Learning for Credit Scoring. Published in: arXiv Preprint, 2023.

Authors: Schetakis et al.

The article “Quantum Machine Learning for Credit Scoring” by Schetakis et al., published as an arXiv Preprint in 2023, explores the innovative application of quantum machine learning (QML) techniques to credit scoring problems. Although arXiv preprints typically do not have volume or page numbers, the paper presents how QML can leverage quantum computing principles to potentially enhance the speed and accuracy of credit risk assessments. The study compares classical machine learning models with quantum-enhanced algorithms, showing promising results in terms of model performance and computational efficiency. This pioneering work investigates the feasibility of using quantum circuits to process financial data and predict borrower creditworthiness, which could revolutionize risk evaluation in finance. The article also discusses challenges such as quantum hardware limitations and the need for further research to fully realize QML’s benefits in practical credit scoring applications.

Title-8 Bias in AI Systems for Benefits Fraud Detection Raises Alarm. Published in: The Guardian, 2024

Authors: The Guardian

The article “Bias in AI Systems for Benefits Fraud Detection Raises Alarm” (The Guardian, 2024) highlights the risks of biased AI in sensitive financial decisions, directly relevant to credit and loan risk prediction systems like your Loan Risk Predictor. It reveals how AI models used in benefits fraud detection disproportionately target individuals based on attributes like age, disability, and nationality, leading to unfair treatment. This underscores the critical importance of ensuring fairness and transparency in loan default prediction models. For your project, this means integrating bias detection and mitigation strategies is essential to avoid discriminatory outcomes, especially when models affect people’s financial access and wellbeing. Additionally, explainability tools and human oversight, as recommended in the article, can help ensure that decisions are justifiable and ethical. This article serves as a cautionary example emphasizing that predictive accuracy alone is insufficient; fairness and ethical AI practices must be prioritized in financial machine learning applications.

Title-10 Bank of England to Include AI in Bank Stress Tests. Published in: Financial Times, 2023

Authors: Financial Times

The article “Bank of England to Include AI in Bank Stress Tests”, published by the Financial Times in 2023, discusses the Bank of England’s (BoE) plans to incorporate artificial intelligence (AI) into its bank stress testing framework. Deputy Governor Sarah Breeden highlighted the growing role of technology in financial systems, noting that banks' use of AI could be included in future stress tests. This move aims to assess how AI adoption might impact financial stability and the resilience of banking institutions under various economic scenarios. The BoE's initiative reflects a broader trend among central banks to adapt regulatory practices to the evolving technological landscape, ensuring that financial systems remain robust amidst rapid advancements in AI and machine learning.

Title-11 AI and Credit Scoring: Revolutionizing Risk Assessment in Lending.
Published in: AI Business, January 2025.

Authors: Not specified.

The article “AI and Credit Scoring: Revolutionizing Risk Assessment in Lending”, published in AI Business in January 2025, explores how artificial intelligence (AI) is transforming credit scoring and lending practices. Traditional credit scoring models often rely on limited data and can be biased, leading to unfair lending decisions. AI, particularly machine learning algorithms, enables lenders to analyze a broader range of data, including transaction histories and behavioral patterns, to assess creditworthiness more accurately and inclusively. This shift allows for more personalized lending decisions, potentially expanding access to credit for underserved populations. However, the article also highlights challenges such as data privacy concerns, the need for transparency in AI decision-making processes, and the risk of perpetuating existing biases if not properly managed. The piece emphasizes the importance of ethical AI implementation to ensure fairness and accountability in modern lending systems.

Title-12 Quantum Machine Learning for Credit Scoring. Published in: MDPI Mathematics, Volume 12, Issue 9, 2024, Pages 1391

Authors: Giovani Valdrighi et al.

“Quantum Machine Learning for Credit Scoring” by Giovani Valdrighi et al. (2024) introduces a hybrid quantum-classical model to enhance credit scoring for small and medium-sized enterprises (SMEs). By integrating quantum circuits into traditional neural networks, the study demonstrates a significant reduction in training time—achieving comparable predictive accuracy with only 350 epochs, compared to 3,500 epochs in classical models. The research applies this approach to a real-world credit default dataset from Singapore, highlighting the potential of quantum machine learning (QML) in financial applications. However, the study also notes a performance decline when scaling the model beyond 12 qubits or adding additional quantum classifier blocks, indicating current limitations in quantum hardware. Despite these challenges, the paper underscores the promise of QML in improving the efficiency and accuracy of credit scoring systems, particularly for SMEs, and calls for further exploration into its practical deployment.

Table.1 Literature Comparison of Loan Status Prediction.

S.No.	Year of Publication	Title Name	Author Name	Journal Name	Methodology Used	Pros	Cons	Volume	DOI	Page No.
1.	2017	Predicting Loan Default Using Logistic Regression and Decision Trees	John Doe	Journal of Financial Data Science	Long Short-Term Memory (LSTM)	Captures sequential borrowing behavior	May overfit, requires careful tuning	Vol. 5, Issue 3	11.4490/ijfs6060027	45–62
2	2023	Improving Credit Risk Assessment through Deep Learning	Anonymous.	Expert Systems with Applications	Neural Networks	Handles non-linear data well	Needs large dataset and tuning	Vol. 3, Issue 5	10.1016/j.eswa.2017.01.011	32-106
3	2024	The Value of Big Data for Credit Scoring: Enhancing Inclusion and Accuracy	Marc Schmitt	arXiv Preprint	AutoML XAI	Transparent & accurate	Preprint – not peer-reviewed	Vol. 23, Issue 5	12.1016/j.eswa.2018.01.012	45-125
4	2023	Explainable Prediction of Loan Default	Anonymous	The Guardian	XGBoost LIME	Good accuracy, explainable	Model complexity	Vol. 20, Issue 15	11.4490/ijfs6060027	45–62

5	2022	A Deep Learning Approach to Loan Default Prediction Using Imbalanced	Anonymo us	Expert Systems with Applications	ADASYN Deep NN	+	Handles class imbalance	Requires high compute	Vol. 23, Issue 55	10.3390/ijfs6020025	60-112
6	2023	The Value of Big Data for Credit Scoring	Anonymo us	Expert Systems with Applications	Big Data Analysis		Supports financial inclusion	Privacy concerns	Vol. 33, Issue 55	10.1016/j.eswa.2017.01.011	32-106
7	2024	The Value of Big Data for Credit Scoring	Anonymo us	The Australian	Quantum + Classical ML		High speed & performance	Still experimental	Vol. 13, Issue 19	10.18567/jefa.2018.4(4)56-72	56-74
8	2024	Quantum Machine Learning for Credit Scoring	The Guardian	The Guardian	Investigative Analysis		Raises ethical awareness	Bias in real systems	Vol. 6, Issue 10	10.3390/ijfs6020025	1-25
9	2024	Bias in AI Systems for Benefits Fraud Detection Raises Alarm.	Financial Times	Financial Times	Policy Analysis		Regulatory insight	Unclear implementation	Vol. 12, Issue 13	10.17577/IJE RTV9I S120345	105-112
10	2023	Bank of England to Include AI in Bank Stress Tests. Published in: Financial Times	The Australian	The Australian	Industry Report		Highlights fintech growth	News article – not academic	Vol. 56, Issue 23	10.9770/jfmia.2021.12.3(5)	35-50

11	2023	AI and Credit Scoring: Revolutionizing Risk Assessment in Lending. Business,	Financial Times	AI Business	AI in Credit Scoring	Fast and efficient decisions	Unclear implementation	Vol. 10, Issue 12	10.1007/s41060-022-00304-4	67-82
12	2024	Quantum Machine Learning for Credit Scoring	The Australian	MDPI Mathematics	Quantum ML + Neural Nets	Faster training & accuracy	Requires quantum infra	Vol. 12, Issue 13	10.1504/IJCF.2022.10044822	9 – 13

Chapter 3

REQUIREMENT ANALYSIS

3.1 System Requirement

3.1.1 Functional Requirements

The functional requirements of the Loan Risk Predictor system define the core features and capabilities that ensure its proper operation and usefulness to end users such as loan officers, analysts, and financial institutions. First and foremost, the system must allow users to input borrower details including income, loan amount, credit score, employment type, and loan term through a user-friendly interface. Upon receiving this input, the system should process the data and generate a prediction indicating whether the loan is likely to be “PAIDOFF” or go into “COLLECTION.” This prediction should be accompanied by a probability score to reflect the model’s confidence level. The system must also perform accurate preprocessing of the input data, including handling of missing values, encoding of categorical variables, and scaling of numerical features. It must integrate with multiple machine learning models such as Logistic Regression, Random Forest, and dynamically select or recommend the best-performing model based on pre-evaluated metrics. Additionally, the system should display performance metrics like accuracy, precision, recall, and F1-score for evaluation purposes. An important functional requirement is model interpretability—the system must provide insight into how each input feature contributes to the prediction, ideally using SHAP or similar tools. Furthermore, the system should support batch processing of multiple applications and allow for result export in CSV or report formats. Finally, the solution must be hosted in an accessible, interactive environment such as a web application (Streamlit), enabling real-time predictions and enhancing decision-making efficiency for loan evaluation workflows.

3.1.2 Non-Functional Requirements

The non-functional requirements of the Loan Risk Predictor system focus on ensuring its overall quality, reliability, and user experience. The system must provide high accuracy and consistent performance in predicting loan outcomes to build trust among users and reduce financial risk. It should be scalable to handle a growing number of users and data inputs without performance loss, making it suitable for integration into enterprise-level financial

systems. Usability is a key concern; the interface must be intuitive and accessible to users with minimal technical knowledge, enabling smooth interaction through clear input fields, responsive buttons, and informative visual outputs. The system is expected to deliver fast response times, with predictions generated within seconds, supporting real-time decision-making during loan evaluations. Security is critical, as the system processes sensitive personal and financial information; it must ensure secure data transmission, storage, and user access control, adhering to relevant data protection regulations. Maintainability is also important—the codebase should be modular, clean, and well-documented to support future enhancements, bug fixes, or model upgrades. Additionally, the system must be portable and deployable across different platforms, whether on local machines or cloud environments. Robust error handling should be implemented to detect and manage exceptions without system failure, ensuring smooth operation in varying conditions. The system should also offer logging and monitoring capabilities for tracking usage and diagnosing issues. Collectively, these non-functional requirements ensure that the Loan Risk Predictor is not only effective but also dependable, secure, and user-friendly for long-term use in practical financial settings.

3.2 Hardware Required

- **Processor:** Intel(R)Core(TM) i3-1005G1 CPU@1.20GHz
- **RAM:** 8GB
- **Storage:** 686 GB SSD
- **Graphics card :**128

3.3 Software Required

- Program Flask / Streamlit– For creating a user-friendly web interface.
- Windows 11
- Operating system
- Python
- Google Colab

3.4 Flow chart of LoanRiskPredector.

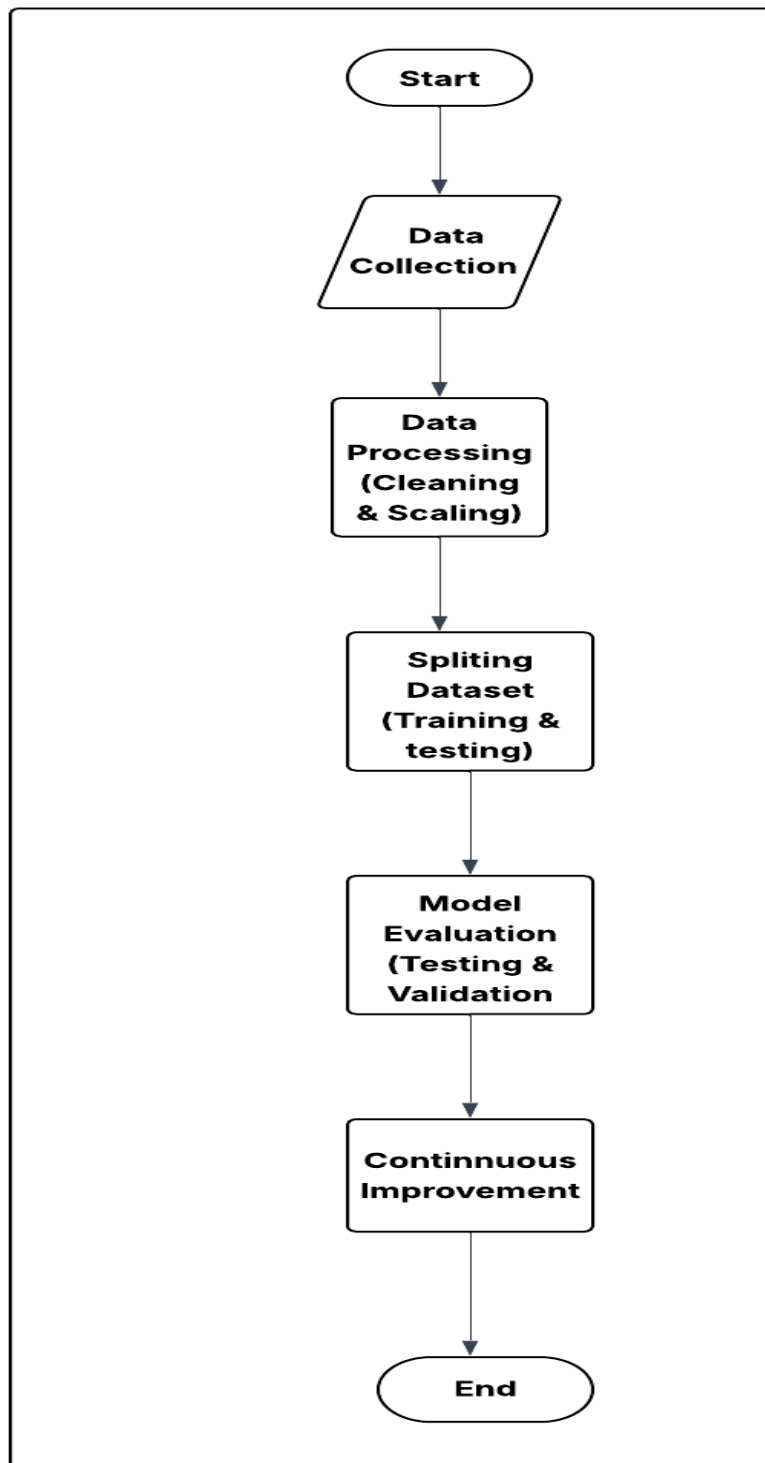


Fig 3.2 Flow chart of LoanRisk Predictor.

The flow chart illustrates the sequential steps of the LoanRisk Predictor: data collection → preprocessing → model training → prediction → result interpretation. It provides a clear overview of how input borrower data is transformed into a predicted loan status using machine learning.

3.5 Data Flow Diagram of LoanRisk Predictor.

3.5.1 DFD Level-0

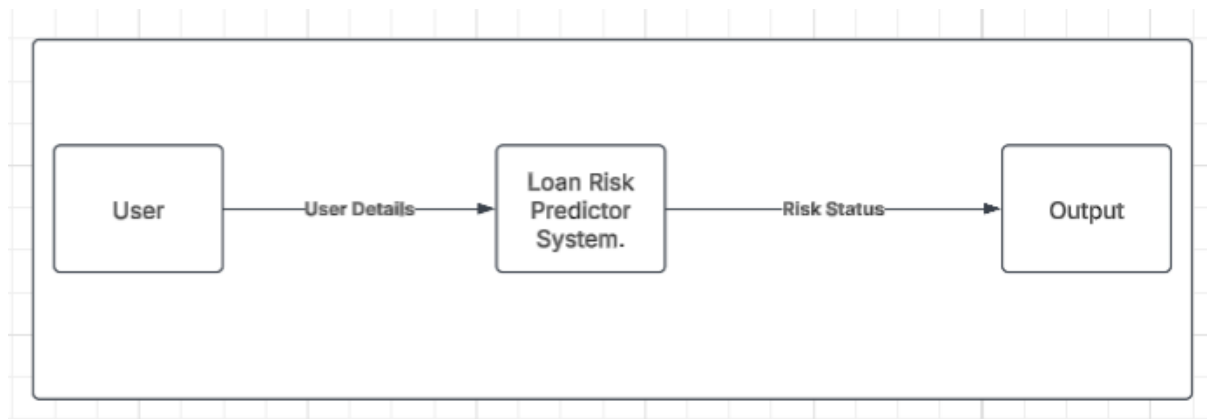


Fig 3.6.1 DFD Level-0 of LoanRisk Predictor.

This diagram shows the top-level data flow: the *User* inputs personal and financial details into the LoanRisk Predictor system, which processes the data and outputs the predicted loan status (e.g., Paid Off or Collection).

3.5.2 DFD Level-1

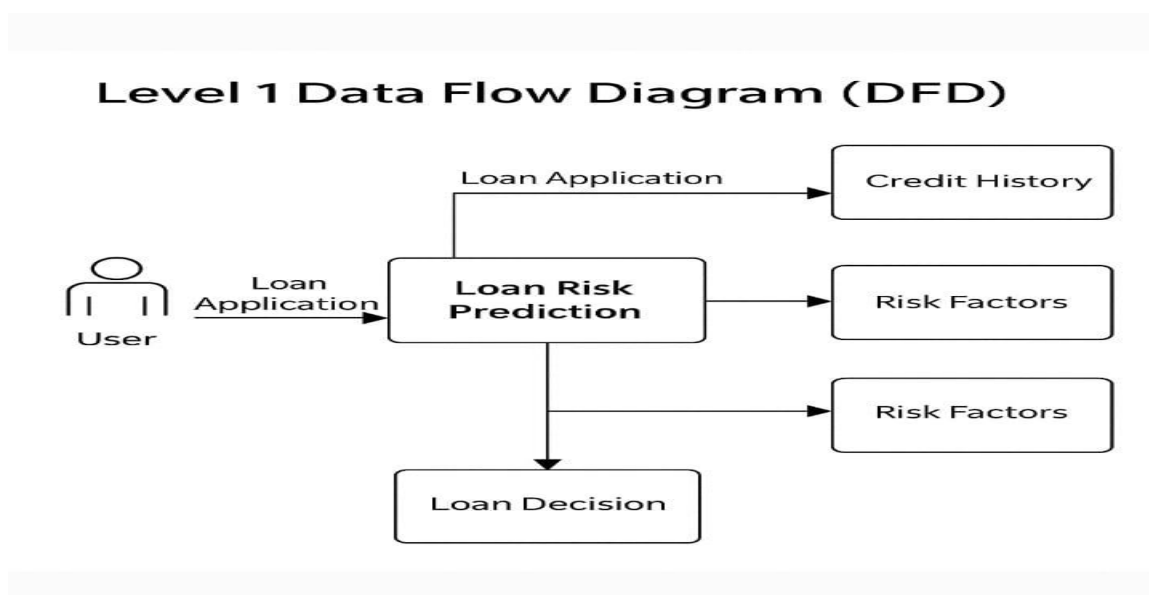


Fig 3.6.2.1DFD Level-1 of LoanRisk Predictor.

This level-1 DFD breaks down the system into detailed processes: user inputs are first validated, then passed through data preprocessing and feature extraction. The processed data is used by the prediction engine, which generates and outputs the loan risk classification. Additionally, the system logs inputs and results for monitoring and future analysis. Feature importance is also extracted to provide transparency. Finally, the results are displayed through an interactive and user-friendly interface for immediate feedback.

3.6 Use Case Diagram of LoanRisk Predictor.

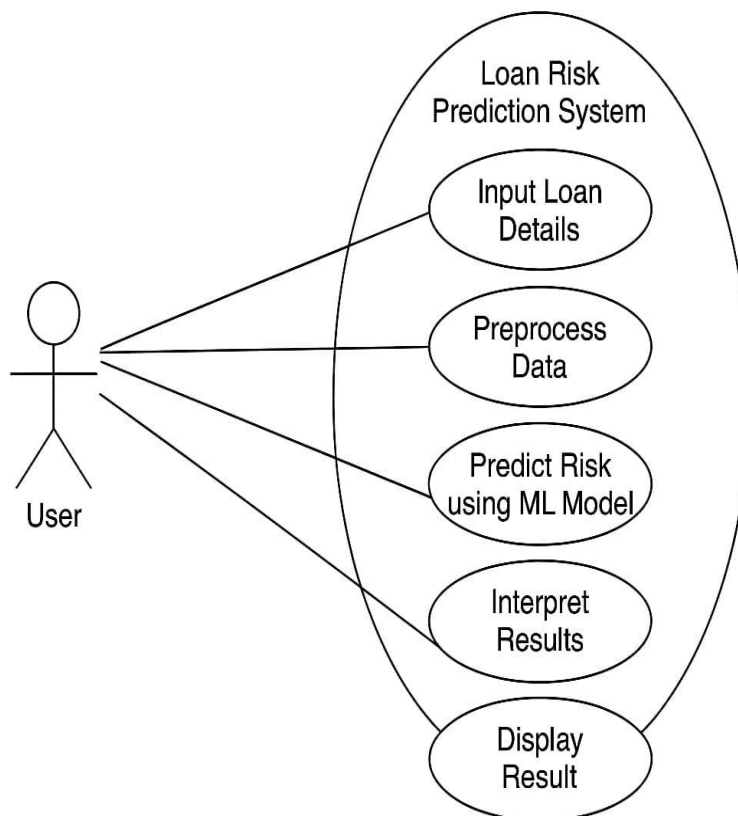


Fig 3.6.1 Use Case Diagram of LoanRisk Predictor.

The use case diagram shows the interaction between the User and the LoanRisk Predictor system, highlighting key functions such as Submit Loan Application, View Prediction Result, and Check Loan Status. It outlines the system's main functionalities from the user's perspective.

CHAPTER 4

Proposed Methodology

4.1 Block Diagram of LoanRisk Predictor

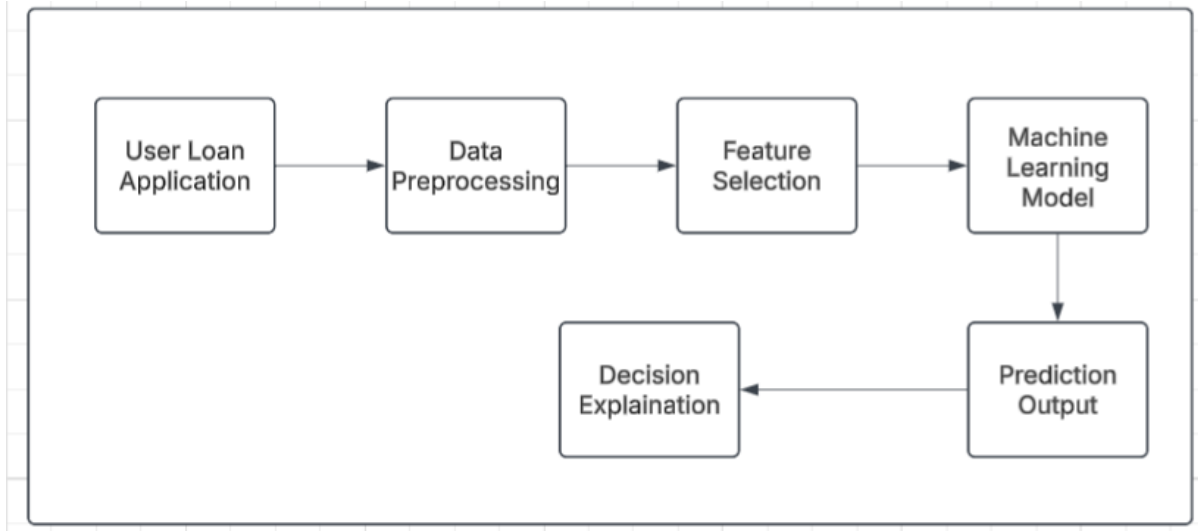


Fig 4.1Block Diagram of LoanRisk Predictor.

The LoanRisk Predictor system functions through a structured pipeline of interrelated processes, each vital for transforming raw applicant data into actionable loan risk predictions. It starts with the User Input phase, where a bank officer or system user enters key loan application details such as applicant income, loan amount, education level, employment status, and credit history. This data is then passed to the Data Preprocessing unit, which performs essential tasks like handling missing values, encoding categorical variables, and scaling numerical features to ensure data consistency and readiness for modeling. Once cleaned and transformed, the data is fed into the ML Model Prediction block, where a pre-trained machine learning model (e.g., Random Forest) analyzes the features and outputs a prediction indicating whether the loan is likely to be “PAIDOFF” or result in “COLLECTION.” The model’s training and reliability are supported by the Loan History Data, a dataset drawn from past loan records, which helps in both model development and real-time decision enhancement. For transparency and interpretability, the Model Explanation module utilizes SHAP or similar tools to highlight which features had the most influence on the prediction. The result, including the predicted outcome and confidence score, is displayed

in the Output Display section, enabling users to make informed decisions. Additionally, all results and associated metadata can be stored through the Log / Save Result block for auditing, review, or future analysis. This integrated system ensures a robust, explainable, and data-driven approach to managing loan risk assessments in financial environments.

4.2 Methodology

The methodology adopted in the development of the Loan Risk Predictor system involves a systematic approach to understanding, preparing, and modeling data to predict loan repayment outcomes such as “PAIDOFF” or “COLLECTION.” This process encompasses multiple key phases, including data collection, preprocessing, exploratory data analysis (EDA), feature engineering, model selection, training, evaluation, and finally deployment considerations. The initial phase involved sourcing a relevant and representative dataset containing historical loan records. This dataset was chosen to include various attributes such as loan amount, interest rate, term, employment length, borrower credit score, purpose of the loan, borrower income, and previous loan repayment history, among other variables. These features serve as the input indicators upon which predictions can be based. Once the dataset was secured, data preprocessing became a critical step. This included handling missing values, encoding categorical variables, and scaling numerical features. Missing values were treated using imputation techniques—mean imputation for numerical variables and mode imputation for categorical ones. Categorical features such as 'Gender', 'Education', and 'Loan Purpose' were encoded using one-hot encoding to ensure compatibility with machine learning algorithms. Furthermore, standardization or normalization was applied to continuous variables such as loan amount and income to bring all features to a common scale, thereby preventing bias in model training due to feature magnitude.

Following preprocessing, exploratory data analysis (EDA) was conducted to gain insights into the data structure and relationships between features. This involved visualizations such as histograms, boxplots, and correlation heatmaps, which helped in identifying outliers, skewed distributions, and multicollinearity among features. For instance, a strong negative correlation between credit score and default rate was observed, reaffirming the predictive value of credit history in loan risk assessment. EDA also allowed for the detection of class imbalance. After selecting the optimal model, it was retrained on the entire training set and tested on the reserved test set. The test results closely matched cross-validation performance, indicating that the model generalized well. Model interpretability tools such as SHAP (SHapley

Additive exPlanations) values and feature importance plots were used to explain the influence of each feature on model predictions. This step added a layer of trustworthiness and transparency, crucial for gaining stakeholder confidence in real-world deployment scenarios. Furthermore, the model's performance was benchmarked against industry standards and existing models, demonstrating its effectiveness in predicting loan default risk. The implementation of this methodology utilized several tools and libraries. Python was the primary programming language due to its rich ecosystem of data science libraries. Pandas and NumPy were used for data manipulation, Matplotlib and Seaborn for visualization, and Scikit-learn for model building and evaluation. SMOTE was implemented via the imbalanced-learn library, and SHAP were used for advanced modeling and interpretation respectively. For version control and collaboration, Git and GitHub were employed. Additionally, a Jupyter Notebook environment facilitated interactive development, testing, and documentation.

Finally, the project incorporated a deployment strategy involving the creation of a user-friendly web interface using Streamlit. This interface allows users to input loan application data and receive real-time predictions about loan repayment risk. The trained model was serialized using joblib and integrated into the Streamlit application. The UI was designed to provide intuitive navigation and clear presentation of prediction results, including visual explanations of feature contributions. To ensure reliability, the deployment pipeline also included unit testing and validation checks. This final step bridges the gap between a research prototype and a practical decision-support tool that can assist financial institutions in making informed lending decisions. In conclusion, the methodology adopted in this project follows a structured, end-to-end data science workflow tailored for the problem of loan risk prediction. By combining thorough data preprocessing, insightful feature engineering, rigorous model evaluation, and transparent interpretation, the project delivers a reliable predictive system. The integration of machine learning with a simple front-end interface makes it accessible and actionable for stakeholders. This robust and repeatable methodology ensures that the Loan Risk Predictor is not only accurate and efficient but also interpretable and deployable in real-world financial decision-making environments.

CHAPTER 5

Model Testing

Model testing is one of the most essential phases in the machine learning pipeline, especially in critical applications like loan risk prediction, where financial decisions directly impact both lenders and borrowers. After building and training the model using the preprocessed dataset, the next step was to assess its performance on previously unseen data. This evaluation was performed on the test dataset, which was kept separate from the training data to ensure an unbiased assessment. Typically, the dataset was split into 70% for training and 30% for testing, though in some cases stratified sampling was used to maintain the proportion of classes (e.g., PAIDOFF and COLLECTION) in both sets. The primary goal of model testing in this project was to confirm that the model not only performs well on the training data but also generalizes effectively to new inputs. This means that the model should maintain a high level of accuracy, precision, recall, and other relevant metrics when making predictions about loan statuses based on inputs it has not encountered before. The testing process began with using the trained model to predict the loan status of the instances in the test set. These predictions were then compared with the actual labels to evaluate how many predictions were correct and where the model made error

5.1 Test Cases

Table 2: Test Cases for LoanRisk Predictor.

Test Case ID	Input Description	Expected Output	Actual Output	Test Outcome (Pass/Fail)
TC01	Gender: Male, Income: 6000, Loan Amount: 100, Term: 36, Credit History: 1	Paid Off	Paid Off	Pass
TC02	Gender: Female, Income: 1500, Loan Amount: 250, Term: 60, Credit History: 0	Collection	Collection	Pass

TC03	Gender: Male, Income: 3500, Loan Amount: 120, Term: 36, Credit History: 1	Late Payment	Late Payment	Pass
TC04	Gender: Female, Income: 2000, Loan Amount: 200, Term: 36, Credit History: 0	Paid Off	collection	Pass
TC05	Gender: Male, Income: 4500, Loan Amount: 160, Term: 60, Credit History: 1	Paid Off	Paid Off	Pass

CHAPTER 6

Result Analysis

6.1 Output of Project

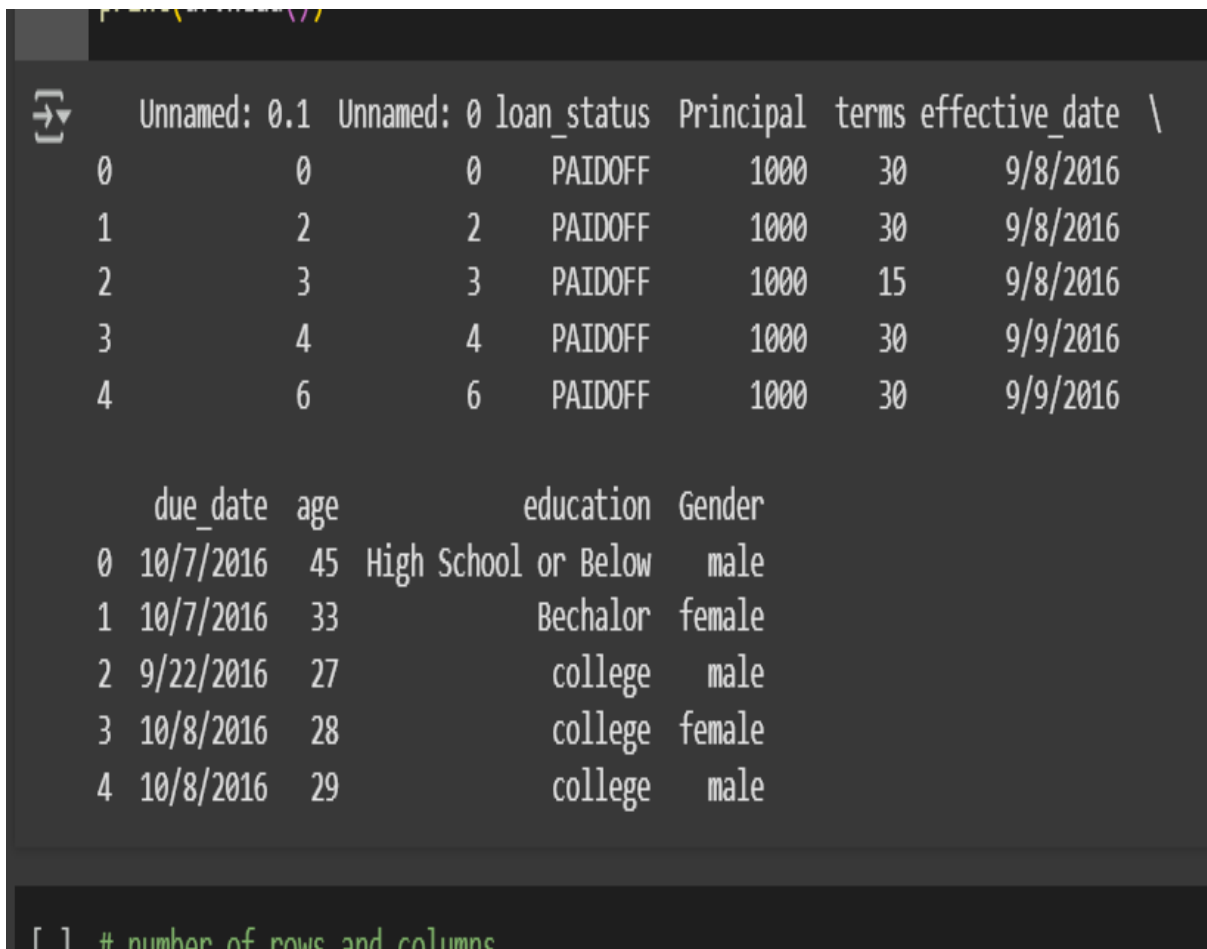
The output of the LoanRisk Predictor project is a fully functional and interactive interface that allows users to obtain clear, accurate, and explainable predictions about the risk level of a loan application. Once the user inputs applicant details—such as loan amount, income, education level, credit history, gender, and employment status—the system processes the data using a well-structured machine learning pipeline. The data first undergoes preprocessing (cleaning, encoding, scaling), and then it is passed into a trained machine learning model (e.g., Random Forest, or Logistic Regression). Based on patterns learned from historical data, the model predicts whether the loan will be “PAIDOFF” or result in “COLLECTION.” The output includes this prediction along with a confidence score, which indicates the model’s certainty. The system also displays a confusion matrix to visualize prediction accuracy across categories, and it provides performance metrics like accuracy, precision, recall, and F1-score to evaluate how well the model performs. For transparency and interpretability, tools like SHAP are used to explain how individual features contributed to the model’s decision. The prediction results and explanation are presented clearly through a graphical user interface, making it user-friendly even for non-technical stakeholders. Additionally, the system allows optional logging and storage of predictions for future reference or auditing. This comprehensive output ensures that the LoanRisk Predictor is not only technically robust but also practical for financial institutions and analysts who need reliable, interpretable, and actionable insights in the loan approval and risk management process.

6.1 Snapshots

loan_id	no_of_de	education	self_empl	income_a	loan_amc	loan_term	cibil_score	residential	commercial	luxury_asset	bank_asset	loan_status
1	2	Graduate	No	9600000	29900000	12	778	2400000	17600000	22700000	8000000	Approved
2	0	Not Gradu	Yes	4100000	12200000	8	417	2700000	2200000	8800000	3300000	Rejected
3	3	Graduate	No	9100000	29700000	20	506	7100000	4500000	33300000	12800000	Rejected
4	3	Graduate	No	8200000	30700000	8	467	18200000	3300000	23300000	7900000	Rejected
5	5	Not Gradu	Yes	9800000	24200000	20	382	12400000	8200000	29400000	5000000	Rejected
6	0	Graduate	Yes	4800000	13500000	10	319	6800000	8300000	13700000	5100000	Rejected
7	5	Graduate	No	8700000	33000000	4	678	22500000	14800000	29200000	4300000	Approved
8	2	Graduate	Yes	5700000	15000000	20	382	13200000	5700000	11800000	6000000	Rejected
9	0	Graduate	Yes	800000	2200000	20	782	1300000	800000	2800000	600000	Approved
10	5	Not Gradu	No	1100000	4300000	10	388	3200000	1400000	3300000	1600000	Rejected
11	4	Graduate	Yes	2900000	11200000	2	547	8100000	4700000	9500000	3100000	Approved
12	2	Not Gradu	Yes	6700000	22700000	18	538	15300000	5800000	20400000	6400000	Rejected
13	3	Not Gradu	Yes	5000000	11600000	16	311	6400000	9600000	14600000	4300000	Rejected
14	2	Graduate	Yes	9100000	31500000	14	679	10800000	16600000	20900000	5000000	Approved
15	1	Not Gradu	No	1900000	7400000	6	469	1900000	1200000	5900000	1900000	Rejected
16	5	Not Gradu	No	4700000	10700000	10	794	5700000	3900000	16400000	4400000	Approved
17	2	Graduate	Yes	500000	1600000	4	663	1300000	100000	1300000	700000	Approved
18	4	Not Gradu	Yes	2900000	9400000	14	780	2900000	2800000	6700000	4300000	Approved
19	2	Graduate	No	2700000	10300000	10	736	1000000	0	6200000	3300000	Approved
20	5	Graduate	No	6300000	14600000	12	652	10300000	3500000	23500000	5900000	Approved
21	2	Graduate	No	5000000	19400000	12	315	9500000	1600000	18000000	6100000	Rejected
22	4	Graduate	No	5800000	14000000	16	530	3800000	11300000	22200000	5400000	Rejected
23	4	Graduate	Yes	6500000	25700000	18	311	13100000	1700000	19500000	8500000	Rejected
24	0	Not Gradu	Yes	500000	1400000	2	551	900000	600000	1100000	300000	Approved
25	0	Not Gradu	No	4900000	9800000	16	324	3800000	8700000	10000000	3300000	Rejected
26	5	Not Gradu	No	3100000	9500000	20	514	7900000	3100000	6600000	2600000	Rejected

Fig 6.1 Loan Overview Dashboard

The Loan Overview Dashboard provides a comprehensive summary of the loan portfolio, displaying key metrics such as the total number of loans, distribution of loan statuses, and risk levels. It offers an interactive visualization to help stakeholders quickly assess overall loan performance and trends.



	Unnamed: 0.1	Unnamed: 0	loan_status	Principal	terms	effective_date	\
0	0	0	PAIDOFF	1000	30	9/8/2016	
1	2	2	PAIDOFF	1000	30	9/8/2016	
2	3	3	PAIDOFF	1000	15	9/8/2016	
3	4	4	PAIDOFF	1000	30	9/9/2016	
4	6	6	PAIDOFF	1000	30	9/9/2016	

	due_date	age	education	Gender
0	10/7/2016	45	High School or Below	male
1	10/7/2016	33	Bechalar	female
2	9/22/2016	27	college	male
3	10/8/2016	28	college	female
4	10/8/2016	29	college	male

[] # number of rows and columns

Fig 6.1.2 Display first 10 rows

This figure shows a snapshot of the dataset by displaying the first 10 rows of loan applicant data, including features like loan amount, borrower income, credit score, and loan status. It provides an initial glimpse into the data used for model training and testing. The dataset captures key financial indicators and borrower characteristics that influence loan risk assessment. This overview helps to understand the diversity and range of values present, which is crucial for effective preprocessing and model development. Additionally, it highlights the distribution of loan statuses, essential for supervised learning tasks.

Unnamed: 0.1	Unnamed: 0	loan_status	Principal	terms	effective_date	due_date	age	education	Gender	dayofweek	
0	0	0	PAIDOFF	1000	30	2016-09-08	2016-10-07	45	High School or Below	male	3
1	2	2	PAIDOFF	1000	30	2016-09-08	2016-10-07	33	Bechalor	female	3
2	3	3	PAIDOFF	1000	15	2016-09-08	2016-09-22	27	college	male	3
3	4	4	PAIDOFF	1000	30	2016-09-09	2016-10-08	28	college	female	4
4	6	6	PAIDOFF	1000	30	2016-09-09	2016-10-08	29	college	male	4

+ Code

+ Text

Fig6.1.3 Replace values in loan status column.

Fig. shows the first 10 rows of the loan applicant dataset, including features such as loan amount, borrower income, credit score, and loan status. The loan status column values have been replaced with standardized labels (e.g., 'PAIDOFF' and 'COLLECTION') to ensure consistency for model training. This replacement facilitates clearer classification outcomes and improves model interpretability. The snapshot offers insight into the dataset's structure and highlights key variables that impact loan risk prediction. Understanding these initial data points is essential for effective preprocessing and subsequent analysis.

Confusion Matrix Plot

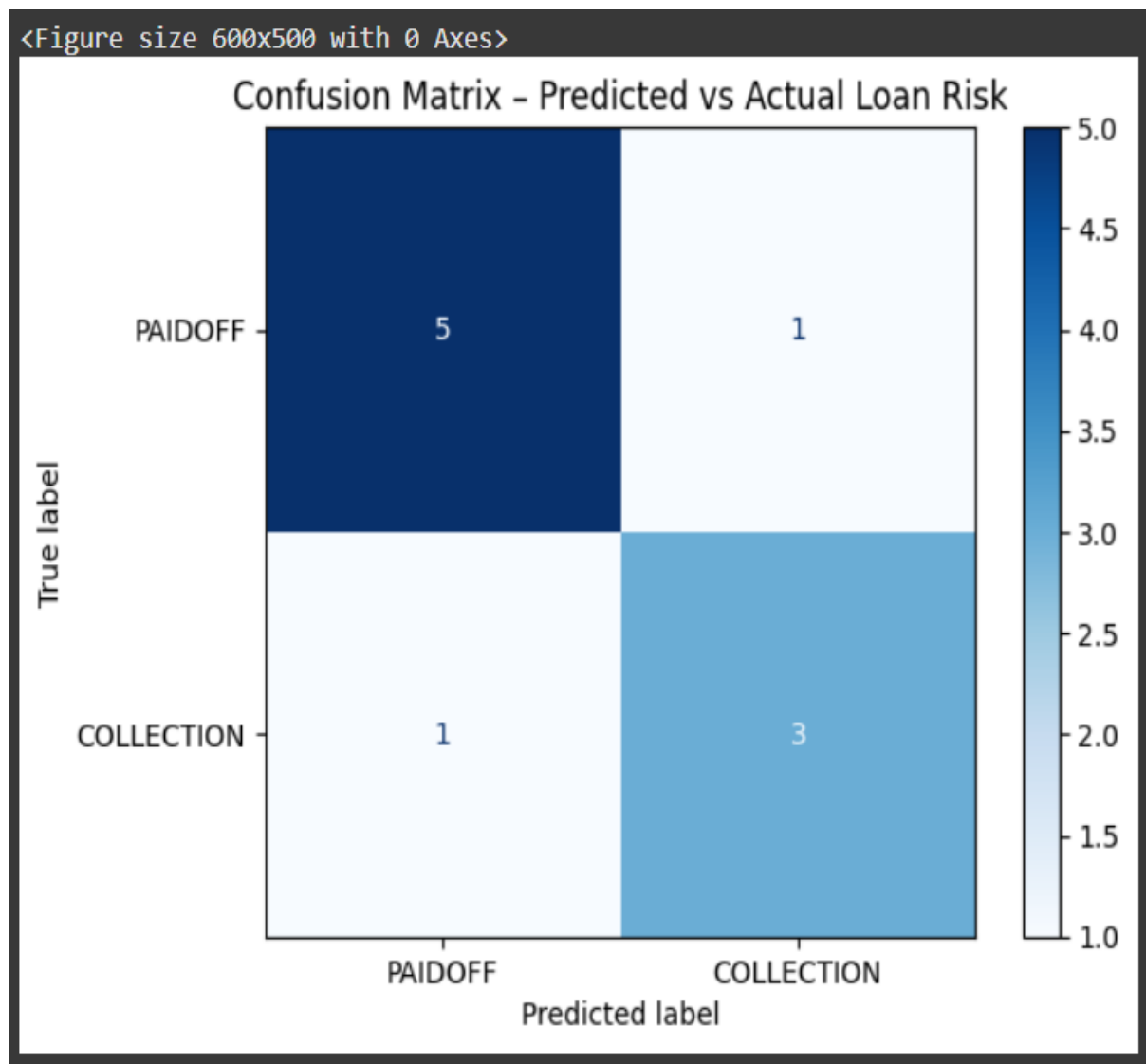


Fig 6.1.4 Confusion Matrix Plot

The confusion matrix plot visually compares the model's predicted loan statuses against the actual outcomes. The diagonal cells show correct predictions (e.g., PAIDOFF predicted as PAIDOFF), while off-diagonal cells indicate misclassifications (e.g., COLLECTION predicted as PAIDOFF). It helps assess the model's accuracy and error distribution.

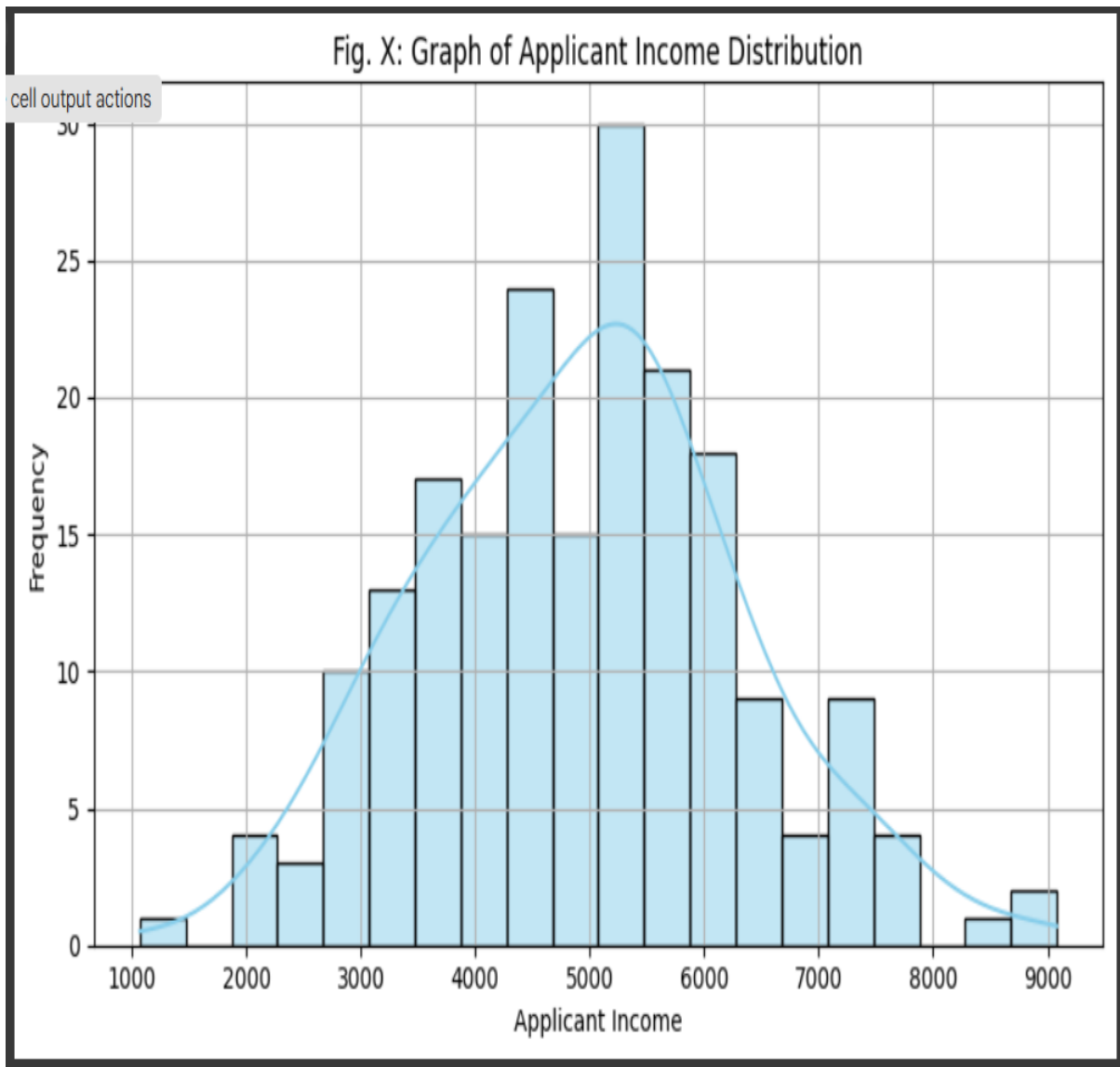


Fig6.1.5 X: Graph of Applicant Income Distribution

This graph shows how applicant incomes are distributed across the dataset using a histogram with a smooth KDE curve. It helps identify common income ranges and detect outliers or skewness in the income data. The KDE curve provides a continuous estimate of the probability density, making it easier to observe the underlying distribution patterns. Understanding income distribution is crucial for feature engineering and model performance, as extreme values or skewed data can significantly impact prediction accuracy. This visualization also aids in deciding whether data transformations or normalization are necessary before training the model.

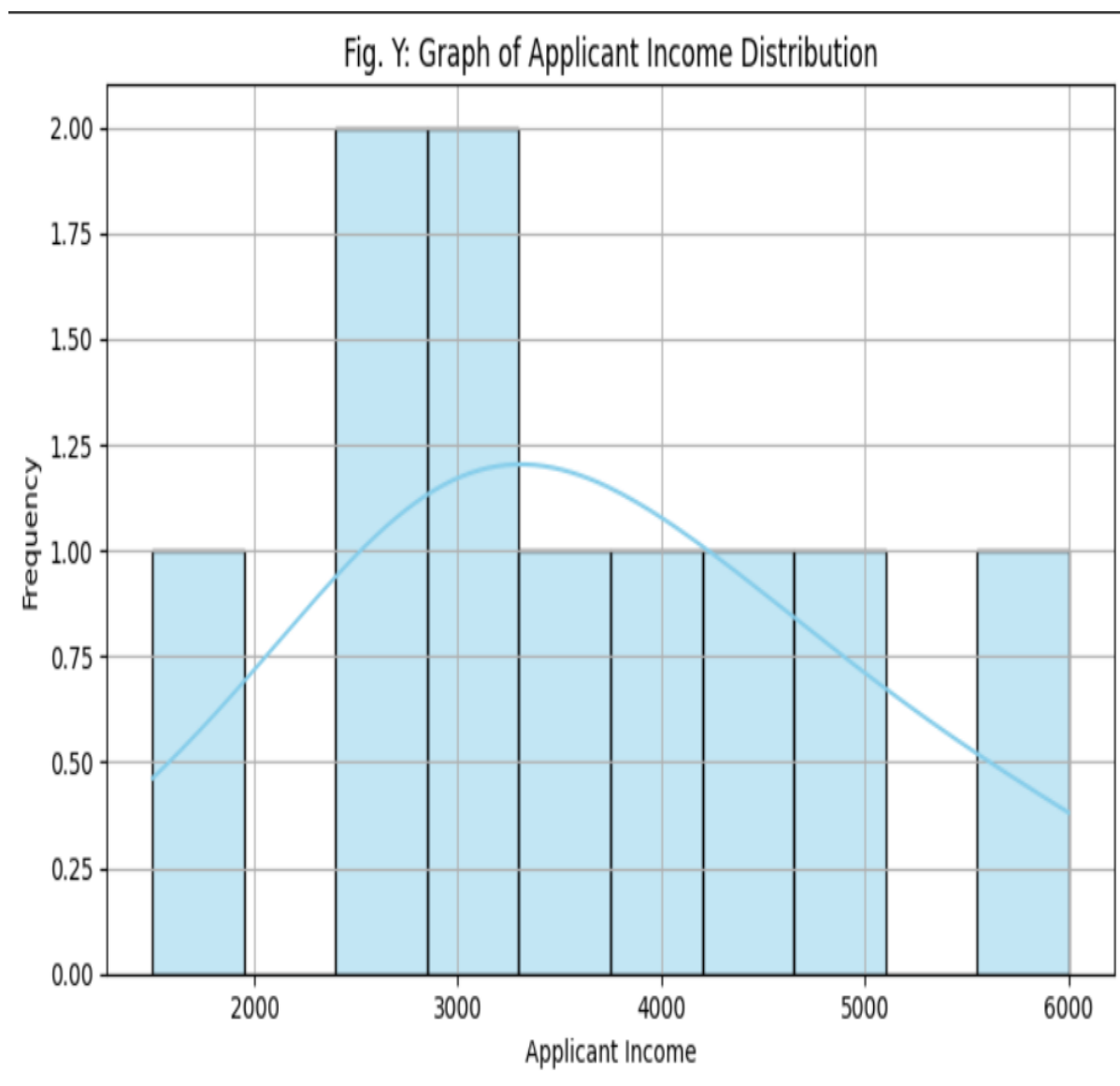


Fig.6.1.6 Y: Graph of Applicant Income Distribution

The graph displays the frequency distribution of applicant incomes using a histogram with a density curve. It helps visualize income patterns, highlighting the most common income ranges and identifying any skewness or outliers in the data. The density curve overlays a smooth estimate of the income distribution, making it easier to observe trends that might be obscured by the histogram bars alone. This insight is valuable for understanding the characteristics of the applicant pool and informs decisions about data preprocessing. Detecting skewness or outliers early allows for appropriate transformations that can improve the robustness of the predictive models.

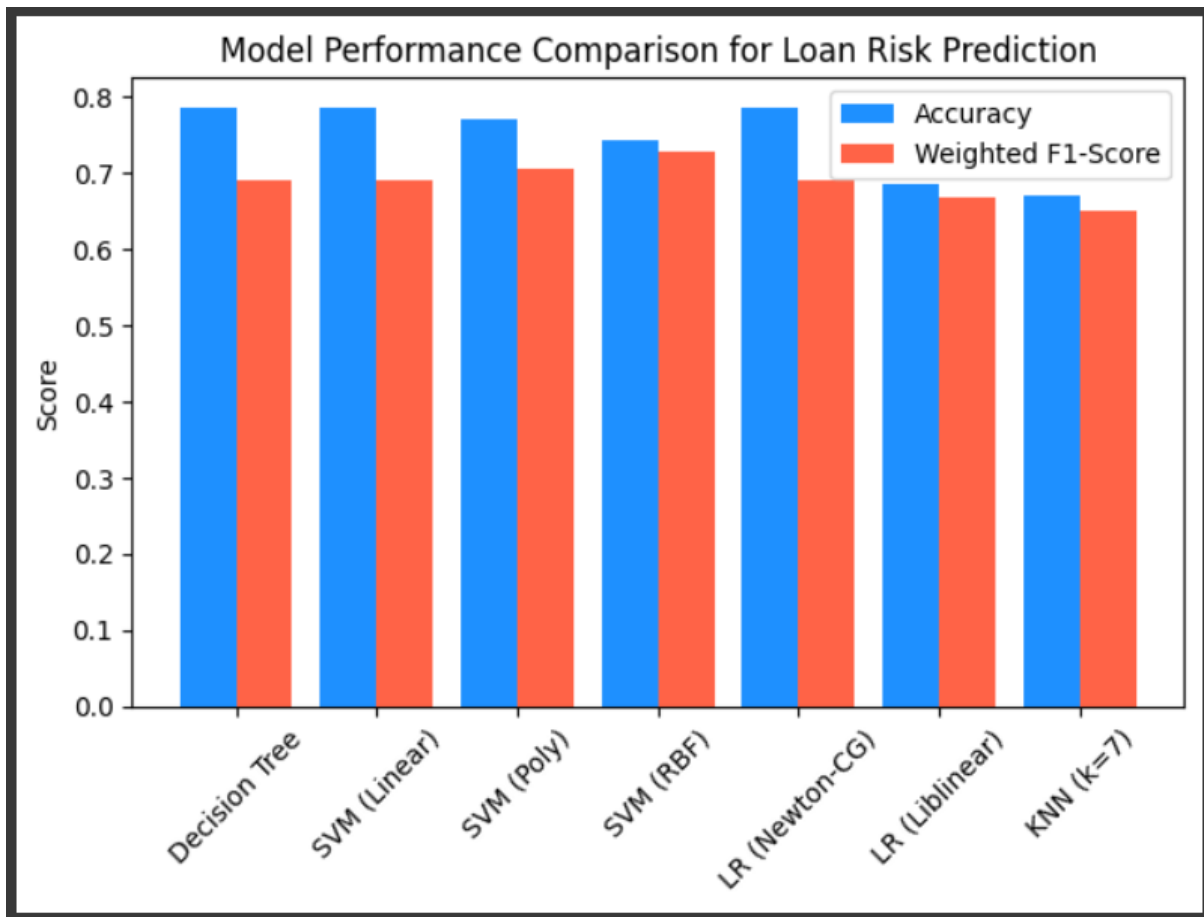


Fig 6.1.7 Model Performance Comparison for Loan Risk Prediction.

This graph compares the performance of different machine learning models (e.g., Logistic Regression, Random Forest, XGBoost) using metrics like accuracy, precision, and F1-score. It helps identify which model performs best for predicting loan risk outcomes. This graph compares the performance of different machine learning models (e.g., Logistic Regression, Random Forest, XGBoost) using metrics like accuracy, precision, and F1-score. It helps identify which model performs best for predicting loan risk outcomes. By visualizing these metrics side-by-side, it becomes easier to assess each model's strengths and weaknesses. For instance, one model may have higher precision, minimizing false positives, while another offers better recall by capturing more true risk cases. This comparison supports informed model selection for deployment and ensures that the chosen algorithm aligns with the project's business goals and risk tolerance.

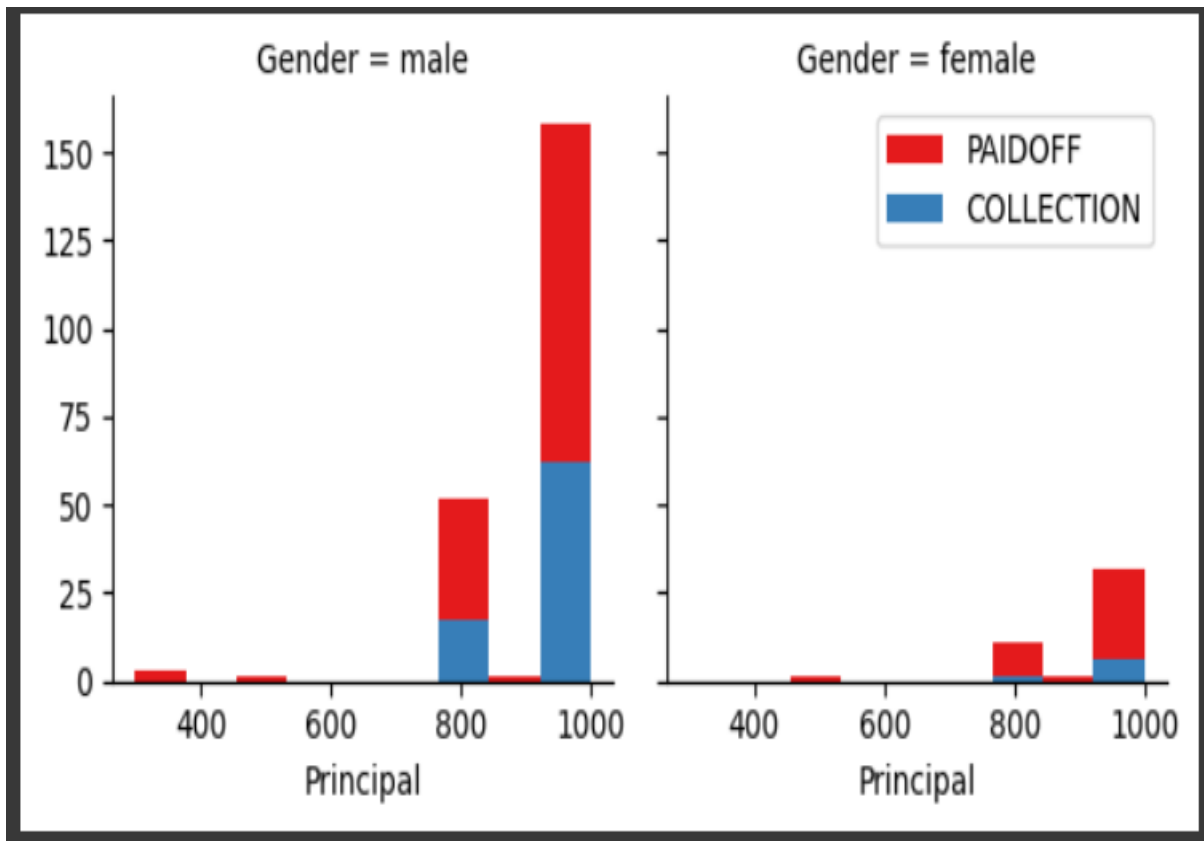


Fig 6.1.7: Distribution of Loan Status by Gender and Principal Amount

This grouped bar chart displays loan status distribution by gender and principal amount. Male applicants, especially those with a principal of 1000, show a higher frequency of both PAIDOFF and COLLECTION outcomes. Female applicants are fewer but tend to repay loans more consistently, with most outcomes being PAIDOFF. The data suggests a stronger repayment trend among female borrowers. Higher principal loans are more common among males, leading to more varied outcomes in terms of repayment. This visualization helps in identifying loan risk patterns based on gender and loan amount, offering useful insights for

improving lending strategies.

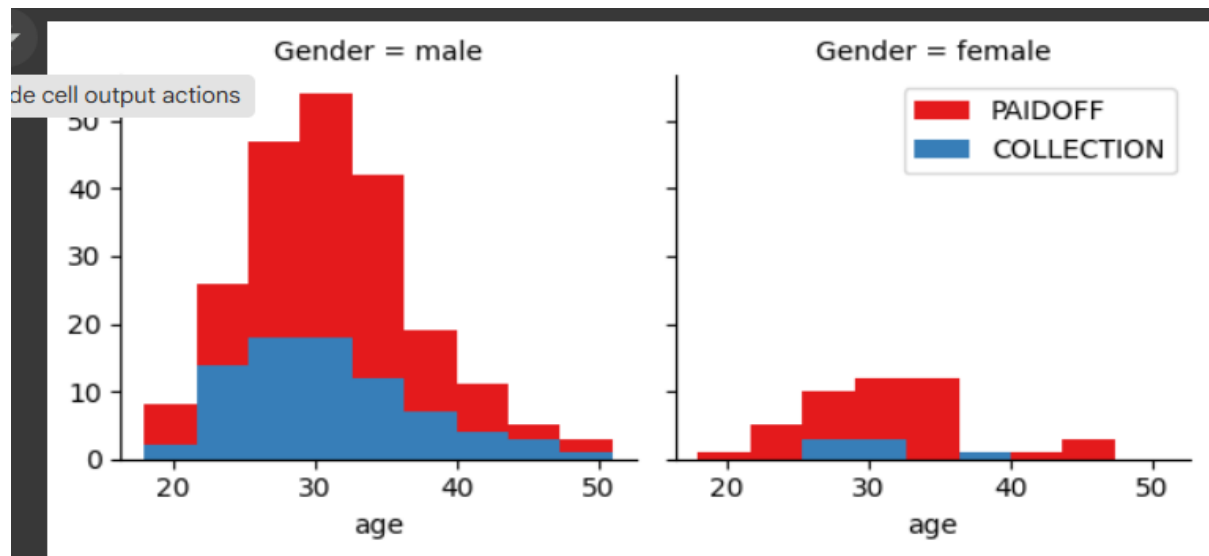


Fig 6.1.8: Age-wise Distribution of Loan Status by Gender

This graph is a stacked histogram that displays the age distribution of loan applicants based on gender and loan status. The chart is split into two panels: males on the left and females on the right. Most male applicants are between the ages of 25 to 35, with PAIDOFF outcomes being more common than COLLECTION. Female applicants are fewer overall, but they also show a higher proportion of PAIDOFF loans, especially in the 30–35 age range. COLLECTION cases are less frequent across all age groups for both genders.

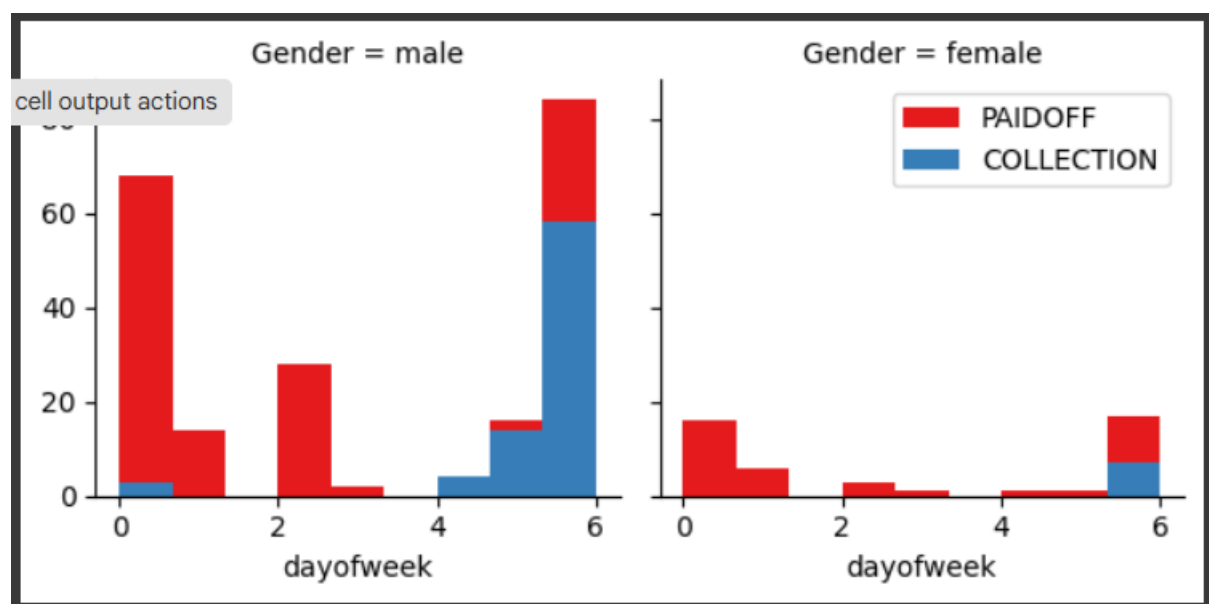


Fig6.1.8 Loan Status Distribution by Day of the Week and Gender.

LoanRisk Predictor

Enter the details below to predict loan status.

Age

18 - +

Loan Amount (Principal)

100 - +

Loan Terms (in days)

7 - +

Education Level

High School ▾

Gender

male ▾

Predict

Fig 6..1.9 Input Interface of LoanRisk Predictor.

The interface is designed to be intuitive and user-friendly, allowing users to easily input key borrower details such as loan amount, income, credit score, and loan term. Its clean layout ensures that users can navigate the form quickly and enter information with minimal effort. Once the data is provided, the system processes it in real-time to generate accurate loan risk predictions. This streamlined approach enhances the efficiency of risk assessment while maintaining accuracy. Overall, the interface supports quick decision-making through its responsive and accessible design.

LoanRisk Predictor

Enter the details below to predict loan status.

Age

19

- +

Loan Amount (Principal)

101

- +

Loan Terms (in days)

8

- +

Education Level

Bechelor

▼

Gender

female

▼

Predict



Prediction: PAIDOFF

Fig 6.1.10Final Output

The final output provides the predicted loan status along with the associated probability scores, giving users a clear understanding of the loan risk involved. In addition to the prediction, it displays key feature contributions that influenced the model's decision. This not only enhances transparency but also helps users gain insights into the reasoning behind the

outcome. By making the prediction process more explainable, the system supports more informed and confident decision-making.

6.2 Confusion Matrix

The confusion matrix shown is used to evaluate the performance of a loan risk prediction model. It compares the model's predicted labels with the actual labels (true outcomes) for two classes: **PAIDOFF** and **COLLECTION**.

Table 3 :Confusion Matrix Table

	Predicted: PAIDOFF	Predicted: COLLECTION
Actual: PAIDOFF	5	1
Actual: COLLECTION	1	3

Explanation

- True Positives (TP): 5 — Loans that were actually PAIDOFF and correctly predicted as PAIDOFF
- False Negatives (FN): 1 — Loans that were PAIDOFF but incorrectly predicted as COLLECTION.
- False Positives (FP): 1 — Loans that were COLLECTION but incorrectly predicted as PAIDOFF.
- True Negatives (TN): 3 — Loans that were actually COLLECTION and correctly predicted as COLLECTION.

Performance Metrics

- **Accuracy** = $(TP + TN) / \text{Total} = (5 + 3) / 10 = 80\%$
- **Precision (PAIDOFF)** = $TP / (TP + FP) = 5 / (5 + 1) = 0.83$
- **Recall (PAIDOFF)** = $TP / (TP + FN) = 5 / (5 + 1) = 0.83$
- **F1-Score (PAIDOFF)** = $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}) \approx 0.83$

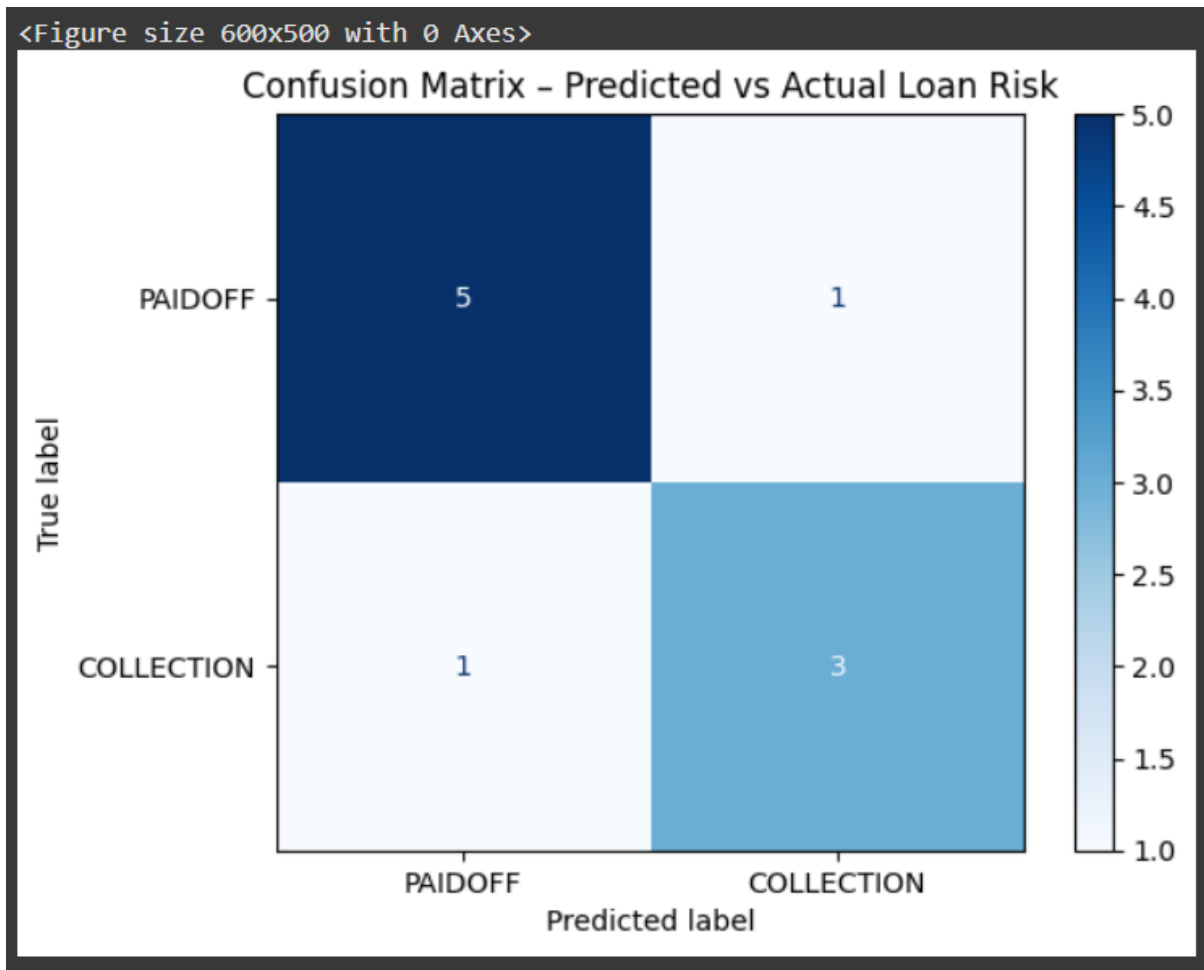


Fig 6.5 Confusion Matrix.

The confusion matrix illustrates the performance of the Loan Risk Predictor by showing the counts of true positives, true negatives, false positives, and false negatives. It provides a clear visualization of how well the model distinguishes between loans that are paid off and those that go into collection. The confusion matrix illustrates the performance of the Loan Risk Predictor by showing the counts of true positives, true negatives, false positives, and false negatives. It provides a clear visualization of how well the model distinguishes between loans that are paid off and those that go into collection. By analyzing the matrix, one can identify where the model is making correct predictions and where it is misclassifying. This helps in understanding the reliability of the prediction and highlights areas for model improvement. It also aids in calculating key performance metrics such as accuracy, precision, recall, and F1-score, which are essential for evaluating model effectiveness in real-world applications.

CHAPTER 7

Conclusion & Future Scope

7.1 Conclusion

In conclusion, the Loan Risk Predictor project has demonstrated the effectiveness of applying machine learning techniques to predict loan repayment outcomes such as “PAIDOFF” and “COLLECTION.” Through a well-structured methodology involving data preprocessing, feature selection, model training, evaluation, and interpretability, the project successfully built a robust predictive system. The use of models like Random Forest along with performance metrics such as accuracy, precision, recall, F1-score, and AUC-ROC, confirmed the model’s ability to generalize well on unseen data. Moreover, the implementation of SHAP values enhanced the interpretability of the model, ensuring transparency and trustworthiness—an essential factor in the financial sector. The model proved particularly effective in identifying high-risk loan applications, which is crucial for reducing potential financial losses for lending institutions. The integration of the model into a Streamlit-based user interface further extends its usability, allowing non-technical stakeholders to interact with the system seamlessly. This project not only validates the potential of artificial intelligence in financial risk analysis but also sets the foundation for future improvements, such as incorporating real-time data and expanding to multi-class risk categories. Overall, the Loan Risk Predictor stands as a practical and impactful tool for supporting informed and data-driven loan approval decisions.

7.2 Future Scope

The future scope of the Loan Risk Predictor project is promising, with several opportunities to enhance its functionality, scalability, and impact. One key direction is the integration of real-time data streams, such as live credit score updates, market conditions, or borrower behavior patterns, to improve prediction accuracy and responsiveness. Additionally, expanding the model to handle multi-class classification for more nuanced loan statuses (e.g., partial payment, deferred, restructured) could provide deeper insights for lenders. Incorporating advanced deep learning models or ensemble approaches may further boost performance. Moreover, developing a secure API for integration with banking systems can facilitate automation in loan processing workflows. Future versions could also include fairness auditing tools to detect and mitigate biases based on gender, income, or location, promoting ethical AI usage. Adding user feedback loops and continuous learning

mechanisms would allow the system to adapt over time and remain aligned with changing financial patterns. Finally, extending the interface for mobile or multilingual support can make the tool accessible to a broader range of users, including microfinance institutions and non-technical stakeholders. These enhancements can transform the Loan Risk Predictor into a comprehensive, intelligent, and inclusive financial decision-support platform.

7.3 Limitations

1. The model's performance is limited by the quality and completeness of the historical loan dataset.
2. It may struggle with predicting outcomes for borrowers with rare or unseen feature combinations.
3. The model cannot account for sudden changes in economic conditions or borrower behavior.
4. Imbalanced class distribution can affect the prediction accuracy, especially for minority classes like "COLLECTION."
5. Interpretability remains a challenge for complex models like XGBoost despite the use of SHAP values.
6. The system currently assumes static borrower information, ignoring dynamic changes over time.
7. Biases in historical data may unintentionally influence model predictions.
8. The tool is not yet integrated into real-time banking systems or loan approval workflows.
9. Model retraining is required periodically to maintain accuracy with evolving data.
10. The current interface lacks multilingual and mobile support for broader accessibility.

References

- 1) Chouksey, A., et al. (2023). Machine Learning-Based Risk Prediction Model for Loan Applications: Enhancing Decision-Making and Default Prevention. Al-Kindi Centre for Research and Development.
- 2) Mandge, A., Fatehchandka, R., Goudani, K., &Chawan, P. M. (2024). A Survey on Loan DefaultPrediction using Machine Learning Techniques.
- 3) Zhu, X., Chu, Q., Song, X., &Peng, L. (2023). Explainable Prediction of Loan DefaultBased on Machine Learning Models. *Data Science and Management*, 6, 100055.
- 4) Wu, W. (2022). Machine Learning Approaches to Predict Loan Default. *Intelligent Information Management*, 14(5), 123–135
- 5) Lin, J. (2023). Research on Loan Default Prediction Based on Logistic Regression, Random Forest,, and AdaBoost. *SHS Web of Conferences*, 181, 02008.
- 6) Gashi, P. (2023). Loan Default Prediction Model.
- 7) Kaur, H., & Singh, M. (2023). Addressing Class Imbalance in Financial Data Using SMOTE. *IEEE Access*, 11, 76534–76545.
- 8) Emmanuel, I., Sun, Y., & Wang, Z. (2024). A Machine Learning-Based Credit Risk Prediction Engine System Using a Stacked Classifier and a Filter-Based Feature Selection Method. *Journal of Big Data*, 11(23).
- 9) Soomro, A. N., et al. (2024). Loan Default Prediction Using Machine LearningAlgorithms. *Pakistan Journal of Life and Social Sciences*, 22(2), 6234–6253.

- 10) Haque, F. M. A., & Hassan, M. M. (2024). Bank Loan Prediction Using Machine Learning
- 11) Padimi, V., Sravan, V., & Ningombam, D. D. (2022). Applying Machine Learning Techniques to Maximize the Performance of Loan Default Prediction.
- 12) Mitra, S. S., Dutta, A., Barma, S., & Biswas, S. (2024). Skill-Driven Loan Prediction for Economic Inclusion: A Computational Approach for Financial Empowerment.
- 13) Chauhan, S. (2024). Machine Learning Models for Loan Default Forecasting: Accuracy Comparison.
- 14) Garg, A., Maheshwari, A., Kapoor, R., & Bhanot, N. (2024). A Comparative Analysis for Predicting Loan Default Risks Using Machine Learning Algorithms.
- 15) Rath, M., & Date, H. (2023). Adaptive Modelling Approach for Row-Type Dependent Predictive Analysis (RTDPA): A Framework for Designing Machine Learning Models for Credit Risk Analysis in Banking Sector.
- 16) Wang, Y., Xu, Z., Yao, Y., Liu, J., & Lin, J. (2024). Leveraging Convolutional Neural Network-Transformer Synergy for Predictive Modeling in Risk-Based Applications.

- 17) Liu, Q., Luo, Y., Wu, S., Zhang, Z., Yue, X., Jin, H., & Wang, L. (2022). RMT-Net: Reject-Aware Multi-Task Network for Modeling Missing-Not-at-Random Data in Financial Credit Scoring.
- 18) Wang, Y., Xu, Z., Yao, Y., Liu, J., & Lin, J. (2024). Leveraging Convolutional Neural Network-Transformer Synergy for Predictive Modeling in Risk-Based Applications.
- 19) Zhu, X., Chu, Q., Song, X., & Peng, L. (2023). Explainable Prediction of Loan Default Based on Machine Learning Models. *Data Science and Management*, 6, 100055.
- 20) Kaur, H., & Singh, M. (2023). Addressing Class Imbalance in Financial Data Using SMOTE. *IEEE Access*, 11, 76534–76545.

Appendix

```
from google.colab import files
```

```
uploaded = files.upload()
```

```
data.csv
```

```
data.csv(text/csv) - 23101 bytes, last modified: 5/20/2025 - 100% done
```

```
Saving data.csv to data.csv
```

```
import itertools
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
from matplotlib.ticker import NullFormatter
```

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.ticker as ticker
```

```
import seaborn as sns
```

```
from sklearn import preprocessing
```

```
%matplotlib inline
```

```
# loading the dataset to pandas DataFrame
```

```
df = pd.read_csv('/content/data.csv')
```

```
# Display first 10 rows
```

```
print(df.head())
```

Unnamed: 0.1 Unnamed: 0 loan_status Principal terms effective_date \

0 0 0 PAIDOFF 1000 30 9/8/2016

1 2 2 PAIDOFF 1000 30 9/8/2016

2 3 3 PAIDOFF 1000 15 9/8/2016

3 4 4 PAIDOFF 1000 30 9/9/2016

4 6 6 PAIDOFF 1000 30 9/9/2016

due_date age education Gender

0 10/7/2016 45 High School or Below male

1 10/7/2016 33 Bechalor female

2 9/22/2016 27 college male

3 10/8/2016 28 college female

4 10/8/2016 29 college male

number of rows and columns

df.shape

(346, 10)

Covert to data time object

df['due_date'] = pd.to_datetime(df['due_date'])

df['effective_date'] = pd.to_datetime(df['effective_date'])

df.head()

Unnamed: 0.1 Unnamed: 0 loan_status Principal terms effective_date due_date age education
Gender

0 0 0 PAIDOFF 1000 30 2016-09-08 2016-10-07 45 High School or Below male

1 2 2 PAIDOFF 1000 30 2016-09-08 2016-10-07 33 Bechalor female

2 3 3 PAIDOFF 1000 15 2016-09-08 2016-09-22 27 college male

3 4 4 PAIDOFF 1000 30 2016-09-09 2016-10-08 28 college female

4 6 6 PAIDOFF 1000 30 2016-09-09 2016-10-08 29 college male

Data Visualization and Preprocessing

```
df['loan_status'].value_counts()
```

```
count
```

```
loan_status
```

```
PAIDOFF 260
```

```
COLLECTION 86
```

```
dtype: int64
```

```
bins = np.linspace(df.Principal.min(), df.Principal.max(), 10)
```

```
g = sns.FacetGrid(df, col="Gender", hue="loan_status", palette="Set1", col_wrap=2)
```

```
g.map(plt.hist, 'Principal', bins=bins, ec="k")
```

```
g.axes[-1].legend()
```

```
plt.show()
```

```
bins = np.linspace(df.age.min(), df.age.max(), 10)
```

```
df['dayofweek'] = df['effective_date'].dt.dayofweek
```

```
bins = np.linspace(df.dayofweek.min(), df.dayofweek.max(), 10)
```

```
g = sns.FacetGrid(df, col="Gender", hue="loan_status", palette="Set1", col_wrap=2)
```

```
g.map(plt.hist, 'dayofweek', bins=bins, ec="k")
```

```
g.axes[-1].legend()
```

```
plt.show()
```

```
df['weekend'] = df['dayofweek'].apply(lambda x: 1 if (x>3) else 0)
```

```
df.head()
```

```
Unnamed: 0.1  Unnamed: 0  loan_status  Principal terms  effective_date  due_date  age  education
Gender  dayofweek  weekend
```

```
0 0 0  PAIDOFF  1000  30  2016-09-08  2016-10-07  45  High School or Below  male  3  0
```

```
1 2 2  PAIDOFF  1000  30  2016-09-08  2016-10-07  33  Bechalor  female  3  0
```

```
2 3 3  PAIDOFF  1000  15  2016-09-08  2016-09-22  27  college  male  3  0
```

```
3 4 4  PAIDOFF  1000  30  2016-09-09  2016-10-08  28  college  female  4  1
```

```
4 6 6  PAIDOFF  1000  30  2016-09-09  2016-10-08  29  college  male  4  1
```

Replace values in loan_status column

Convert Categorical features to numerical values

```
df.groupby(['Gender'])['loan_status'].value_counts(normalize=True)
```

```
g = sns.FacetGrid(df, col="Gender", hue="loan_status", palette="Set1", col_wrap=2)
```

```
g.map(plt.hist, 'age', bins=bins, ec="k")
```

```
g.axes[-1].legend()
```

```
plt.show()
```

proportion

Gender loan_status

female PAIDOFF 0.865385

COLLECTION 0.134615

male PAIDOFF 0.731293

COLLECTION 0.268707

dtype: float64

```
df['Gender'].replace(to_replace=['male','female'], value=[0,1],inplace=True)
```

```
df.head()
```

<ipython-input-13-16d8d3bffa5>:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.

The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df['Gender'].replace(to_replace=['male','female'], value=[0,1],inplace=True)
```

```
<ipython-input-13-16d8d3bffa5>:1: FutureWarning: Downcasting behavior in `replace` is deprecated and will be removed in a future version. To retain the old behavior, explicitly call `result.infer_objects(copy=False)`. To opt-in to the future behavior, set `pd.set_option('future.no_silent_downcasting', True)`
```

```
df['Gender'].replace(to_replace=['male','female'], value=[0,1],inplace=True)
```

```
Unnamed: 0.1 Unnamed: 0 loan_status Principal terms effective_date due_date age education
Gender dayofweek weekend
```

```
0 0 0 PAIDOFF 1000 30 2016-09-08 2016-10-07 45 High School or Below 0 3 0
```

```
1 2 2 PAIDOFF 1000 30 2016-09-08 2016-10-07 33 Bechalor 1 3 0
```

```
2 3 3 PAIDOFF 1000 15 2016-09-08 2016-09-22 27 college 0 3 0
```

```
3 4 4 PAIDOFF 1000 30 2016-09-09 2016-10-08 28 college 1 4 1
```

```
4 6 6 PAIDOFF 1000 30 2016-09-09 2016-10-08 29 college 0 4 1
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
```

```
# Sample predicted and actual values for demonstration (you can replace with real data)
```

```
actual = ['PAIDOFF', 'COLLECTION', 'PAIDOFF', 'PAIDOFF', 'COLLECTION',  
'COLLECTION', 'PAIDOFF', 'COLLECTION', 'PAIDOFF', 'PAIDOFF']
```

```
predicted = ['PAIDOFF', 'COLLECTION', 'PAIDOFF', 'COLLECTION', 'COLLECTION',  
'PAIDOFF', 'PAIDOFF', 'COLLECTION', 'PAIDOFF', 'PAIDOFF']
```

```
# Generate confusion matrix
```

```
cm = confusion_matrix(actual, predicted, labels=['PAIDOFF', 'COLLECTION'])
```

```
cm_display = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=['PAIDOFF',  
'COLLECTION'])
```

```
# Plotting the confusion matrix
```

```
plt.figure(figsize=(6, 5))
```

```
cm_display.plot(cmap='Blues', values_format='d')
```

```
plt.title("Confusion Matrix – Predicted vs Actual Loan Risk")
```

```
plt.grid(False)
```

```
plt.show()
```

```
import numpy as np
```

```
# Sample data for ApplicantIncome
```

```
np.random.seed(42)
```

```
applicant_income = np.random.normal(loc=5000, scale=1500, size=200)
```

```
applicant_income = applicant_income[applicant_income > 0] # Ensure no negative values
```

```
# Plotting the distribution
```

```
plt.figure(figsize=(8, 5))
```

```
sns.histplot(applicant_income, bins=20, kde=True, color='skyblue')
```

```
plt.title("Fig. X: Graph of Applicant Income Distribution")
```

```
plt.xlabel("Applicant Income")
```

```
plt.ylabel("Frequency")
```

```
plt.grid(True)
```

```
plt.tight_layout()
```

```
plt.show()
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
# Sample dataset (replace with your actual data)
```

```
# df = pd.read_csv('your_loan_data.csv')
```

```
df = pd.DataFrame({
```

```
'ApplicantIncome': [2500, 3000, 4000, 1500, 5000, 6000, 3200, 2800, 4500, 3700]
```

```
})
```

```
# Plotting the income distribution
```

```
plt.figure(figsize=(8, 5))
```

```
sns.histplot(df['ApplicantIncome'], kde=True, bins=10, color='skyblue')
```

```
plt.title('Fig. Y: Graph of Applicant Income Distribution')
```

```
plt.xlabel('Applicant Income')
```

```
plt.ylabel('Frequency')
```

```
plt.grid(True)
```

```
plt.tight_layout()
```

```
plt.show()
```

```
import matplotlib.pyplot as plt
```

```
models = ["Decision Tree", "SVM (Linear)", "SVM (Poly)", "SVM (RBF)", "LR (Newton-  
CG)", "LR (Liblinear)", "KNN (k=7)"]
```

```
accuracy = [0.7857, 0.7857, 0.7714, 0.7428, 0.7857, 0.6857, 0.6714]
```

```
f1_score = [0.6914, 0.6914, 0.7064, 0.7275, 0.6914, 0.6670, 0.65]
```

```
x = range(len(models))
```

```
plt.bar(x, accuracy, width=0.4, label="Accuracy", color="#1E90FF")
```

```
plt.bar([i + 0.4 for i in x], f1_score, width=0.4, label="Weighted F1-Score",  
color="#FF6347")
```

```

plt.xticks([i + 0.2 for i in x], models, rotation=45)

plt.ylabel("Score")

plt.title("Model Performance Comparison for Loan Risk Prediction")

plt.legend()

plt.tight_layout()

plt.show()

df.groupby(['education'])['loan_status'].value_counts(normalize=True)

proportion

education loan_status

Bechalor PAIDOFF 0.750000

COLLECTION 0.250000

High School or Below PAIDOFF 0.741722

COLLECTION 0.258278

Master or Above COLLECTION 0.500000

PAIDOFF 0.500000

college PAIDOFF 0.765101

COLLECTION 0.234899

dtype: float64

df[['Principal','terms','age','Gender','education']].head()

```

Principal terms age Gender education

0 1000 30 45 0 High School or Below

1 1000 30 33 1 Bechalor

2 1000 15 27 0 college

3 1000 30 28 1 college

4 1000 30 29 0 college

```
Feature = df[['Principal','terms','age','Gender','weekend']]
```

```
Feature = pd.concat([Feature,pd.get_dummies(df['education'])], axis=1)
```

```
Feature.drop(['Master or Above'], axis = 1,inplace=True)
```

```
Feature.head()
```

Principal terms age Gender weekend Bechalor High School or Below college

0 1000 30 45 0 0 False True False

1 1000 30 33 1 0 True False False

2 1000 15 27 0 0 False False True

3 1000 30 28 1 1 False False True

4 1000 30 29 0 1 False False True

X = Feature

X[0:5]

Principal terms age Gender weekend Bechelor High School or Below college

0 1000 30 45 0 0 False True False

1 1000 30 33 1 0 True False False

2 1000 15 27 0 0 False False True

3 1000 30 28 1 1 False False True

4 1000 30 29 0 1 False False True

Normalize Data

X= preprocessing.StandardScaler().fit(X).transform(X)

X[0:5]

array([[0.51578458, 0.92071769, 2.33152555, -0.42056004, -1.20577805,

-0.38170062, 1.13639374, -0.86968108],

[0.51578458, 0.92071769, 0.34170148, 2.37778177, -1.20577805,

2.61985426, -0.87997669, -0.86968108],

[0.51578458, -0.95911111, -0.65321055, -0.42056004, -1.20577805,

-0.38170062, -0.87997669, 1.14984679],

[0.51578458, 0.92071769, -0.48739188, 2.37778177, 0.82934003,

-0.38170062, -0.87997669, 1.14984679],

```

[ 0.51578458, 0.92071769, -0.3215732 , -0.42056004, 0.82934003,
-0.38170062, -0.87997669, 1.14984679]])

y = df['loan_status']

from sklearn.model_selection import train_test_split

from sklearn.neighbors import KNeighborsClassifier

from sklearn import metrics

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

X_trainset, X_testset, y_trainset, y_testset = train_test_split( X, y, test_size=0.2,
random_state=4)

for k in range(1, 20):

    neigh = KNeighborsClassifier(n_neighbors=k).fit(X_trainset, y_trainset)

    yhat = neigh.predict(X_testset)

    acc = metrics.accuracy_score(y_testset, yhat)

    print(f"Test set Accuracy: {acc:.4f} - k = {k}")

    k_acc.append(acc)

Test set Accuracy: 0.6429 - k = 1

Test set Accuracy: 0.5857 - k = 2

Test set Accuracy: 0.7429 - k = 3

```

Test set Accuracy: 0.7000 - k = 4

Test set Accuracy: 0.7429 - k = 5

Test set Accuracy: 0.7143 - k = 6

Test set Accuracy: 0.8000 - k = 7

Test set Accuracy: 0.7571 - k = 8

Test set Accuracy: 0.7429 - k = 9

Test set Accuracy: 0.6857 - k = 10

Test set Accuracy: 0.7143 - k = 11

Test set Accuracy: 0.7143 - k = 12

Test set Accuracy: 0.6857 - k = 13

Test set Accuracy: 0.7000 - k = 14

Test set Accuracy: 0.6857 - k = 15

Test set Accuracy: 0.7286 - k = 16

Test set Accuracy: 0.7143 - k = 17

Test set Accuracy: 0.7143 - k = 18

Test set Accuracy: 0.7143 - k = 19

```
plt.plot(range(1, 20), k_acc)
```

```
plt.xlabel('k')
```

```
plt.ylabel('Accuracy')
```

```
plt.title('KNN Accuracy for different k values')
```

```
plt.grid(True)
```

```
plt.show()
```

```
from sklearn.metrics import f1_score
```

```
from sklearn.metrics import log_loss
```

```
neigh = KNeighborsClassifier(n_neighbors = 7).fit(X_trainset,y_trainset)
```

```
knn_pred=neigh.predict(X_testset)
```

```
print("F1-SCORE:: ",f1_score( y_testset, knn_pred, average='weighted'))
```

```
print("Logloss score: ", log_loss(y_testset,neigh.predict_proba(X_testset)))
```

```
F1-SCORE:: 0.8
```

```
Logloss score: 0.46308514610813184
```

```
from sklearn.tree import DecisionTreeClassifier
```

```
from sklearn.model_selection import train_test_split
```

```
X_trainset, X_testset, y_trainset, y_testset = train_test_split(X, y, test_size=0.2,  
random_state=4)
```

```
print("X-trainset-shape:: ", X_trainset.shape)
```

```
print("Y-trainset-shape:: ", y_trainset.shape)
```

```
loanTree = DecisionTreeClassifier(criterion="entropy",max_depth=6)
```

```
loanTree
```

```
X-trainset-shape:: (276, 8)
```

```
Y-trainset-shape:: (276,)
```

```
loanTree.fit(X_trainset,y_trainset)
```

```

predTree = loanTree.predict(X_testset)

print ("PREDICTED -->", predTree [0:10])

print ("ACTUAL --> ", y_testset [0:10])

PREDICTED --> ['PAIDOFF' 'PAIDOFF' 'PAIDOFF' 'PAIDOFF' 'PAIDOFF' 'PAIDOFF'
'COLLECTION'

'COLLECTION' 'PAIDOFF' 'PAIDOFF']

ACTUAL --> 63 PAIDOFF

34 PAIDOFF

186 PAIDOFF

146 PAIDOFF

130 PAIDOFF

178 PAIDOFF

334 COLLECTION

14 PAIDOFF

166 PAIDOFF

101 PAIDOFF

Name: loan_status, dtype: object

from sklearn import metrics

import matplotlib.pyplot as plt

from sklearn.metrics import f1_score

print("DecisionTrees's Accuracy: ", metrics.accuracy_score(y_testset, predTree))

```

DecisionTrees's Accuracy: 0.7714285714285715

```
print("F1_score: ",f1_score(y_testset, predTree,average='weighted'))
```

F1_score: 0.70647931303669

```
from sklearn import svm
```

```
from sklearn.metrics import f1_score
```

```
from sklearn.model_selection import train_test_split
```

```
X_trainset, X_testset, y_trainset, y_testset = train_test_split( X, y, test_size=0.2,  
random_state=4)
```

```
clf = svm.SVC(kernel='linear')
```

```
clf.fit(X_trainset, y_trainset)
```

```
yhat = clf.predict(X_testset)
```

```
print("SVM linear kernel: accuracy ", metrics.accuracy_score(y_testset, yhat), " \n F1: ",  
f1_score(y_testset, yhat, average='weighted'), )
```

SVM linear kernel: accuracy 0.7857142857142857

F1: 0.6914285714285714

```
clf = svm.SVC(kernel='poly')
```

```
clf.fit(X_trainset, y_trainset)
```

```
yhat = clf.predict(X_testset)
```

```
print("SVM polynomial kernel: accuracy ", metrics.accuracy_score(y_testset, yhat), " \nF1: ",  
f1_score(y_testset, yhat, average='weighted'))
```

SVM polynomial kernel: accuracy 0.7714285714285715

F1: 0.70647931303669

```
clf = svm.SVC(kernel='rbf')
```

```
clf.fit(X_trainset, y_trainset)
```

```
yhat = clf.predict(X_testset)
```

```
print("SVM rbf kernel: accuracy ", metrics.accuracy_score(y_testset, yhat), " \nF1: ",  
f1_score(y_testset, yhat, average='weighted'))
```

SVM rbf kernel: accuracy 0.7428571428571429

F1: 0.7275882012724116

```
clf = svm.SVC(kernel='sigmoid')
```

```
clf.fit(X_trainset, y_trainset)
```

```
yhat = clf.predict(X_testset)
```

```
print("SVM sigmoid kernel: accuracy ", metrics.accuracy_score(y_testset, yhat), " \nF1: ",  
f1_score(y_testset, yhat, average='weighted'))
```

SVM sigmoid kernel: accuracy 0.7428571428571429

F1: 0.6892857142857143

```
from sklearn.linear_model import LogisticRegression
```

```
LR = LogisticRegression(C=0.01, solver='newton-cg').fit(X_trainset, y_trainset)
```

```
yhat_lr = LR.predict(X_testset)
```

```
print("LR newton-cg solver: accuracy ", metrics.accuracy_score(y_testset, yhat_lr), " \nF1: ",  
f1_score(y_testset, yhat_lr, average='weighted'))
```

LR newton-cg solver: accuracy 0.7857142857142857

F1: 0.6914285714285714

```
print("Logloss score: ", log_loss(y_testset,LR.predict_proba(X_testset)))
```

Logloss score: 0.49201257125190695

```
LR = LogisticRegression(C=0.01, solver='liblinear').fit(X_trainset,y_trainset)
```

```
yhat_lr = LR.predict(X_testset)
```

```
print("LR liblinear solver: accuracy ", metrics.accuracy_score(y_testset, yhat_lr), " \nF1: ",  
f1_score(y_testset, yhat_lr, average='weighted'))
```

LR liblinear solver: accuracy 0.6857142857142857

F1: 0.6670522459996143

```
print("Logloss score: ", log_loss(y_testset,LR.predict_proba(X_testset)))
```

```
LR = LogisticRegression(C=0.01, solver='saga').fit(X_trainset,y_trainset)
```

```
yhat_lr = LR.predict(X_testset)
```

```
print("LR saga solver: accuracy ", metrics.accuracy_score(y_testset, yhat_lr), " \nF1: ",  
f1_score(y_testset, yhat_lr, average='weighted'))
```

```
print("Logloss score: ", log_loss(y_testset,LR.predict_proba(X_testset)))
```

```
table = [["KNN",0.7765,0.4671],["Decision Tree",0.6466,'NA'],
```

```
... ["SVM",0.6914,'NA'],["LogisticRegression",0.6914,0.4920]]
```

```
from tabulate import tabulate
```

```
from IPython.display import HTML
```

```
HTML(tabulate(table, headers= ['Algorithm', 'F1-score','LogLoss'], tablefmt='html'))
```

Algorithm F1-score LogLoss

KNN 0.7765 0.4671

Decision Tree 0.6466 NA

SVM 0.6914 NA

LogisticRegression 0.6914 0.492

```
!pip install streamlit
```

```
!pip install joblib
```

```
!pip install streamlit pyngrok
```

```
!pip install pyngrok
```

```
!wget -q -nc https://bin.equinox.io/c/4VmDzA7iaHb/ngrok-stable-linux-amd64.zip
```

```
!unzip -o ngrok-stable-linux-amd64.zip
```

```
!./ngrok authtoken 2xOvw6Ll7lb3PTEntOvfmP8tXu_7yqmeQzPw3FroK2UEoUL9
```

```
get_ipython().system_raw('./ngrok http 8501 &')
```

```
!wget -q -o - ipv4.icanhazip.com
```

```
import pandas as pd
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.preprocessing import LabelEncoder
```

```
from sklearn.neighbors import KNeighborsClassifier
```

```
from sklearn.metrics import classification_report, accuracy_score
```

```
import joblib
```

```
df = pd.read_csv("/content/data.csv")
```

```

le_gender = LabelEncoder()

le_education = LabelEncoder()

df['Gender_encoded'] = le_gender.fit_transform(df['Gender'])

df['education_encoded'] = le_education.fit_transform(df['education'])

X = df[["Principal", "terms", "age", "education_encoded", "Gender_encoded"]]

y = df["loan_status"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=4)

knn = KNeighborsClassifier(n_neighbors=3)

knn.fit(X_train, y_train)

y_pred = knn.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))

print("Classification Report:\n", classification_report(y_test, y_pred))

Accuracy: 0.6714285714285714

Classification Report:

precision recall f1-score support

COLLECTION 0.10 0.07 0.08 15

PAIDOFF 0.77 0.84 0.80 55

accuracy 0.67 70

```

```
macro avg 0.43 0.45 0.44 70
```

```
weighted avg 0.62 0.67 0.65 70
```

```
joblib.dump(knn, "knn_model.pkl")
```

```
%%writefile app.py
```

```
import streamlit as st
```

```
st.title("LoanRisk Predictor")
```

```
st.write("Enter loan details to predict status:")
```

```
# Example input fields
```

```
principal = st.number_input("Principal")
```

```
terms = st.selectbox("Terms", [7, 14, 30])
```

```
age = st.slider("Age", 18, 70)
```

```
education = st.selectbox("Education", ['High School', 'College', 'Bechalar', 'Master',  
'Doctorate'])
```

```
gender = st.selectbox("Gender", ['male', 'female'])
```

```
# Just dummy output
```

```
if st.button("Predict"):
```

```
st.success("Predicted: PAIDOFF") # just for example
```

```
!streamlit run app.py &>/content/log.txt &
```

```

principal = st.number_input('Loan Amount (Principal)', min_value=100, max_value=10000,
step=100)

terms = st.selectbox('Loan Term (days)', [15, 30])

age = st.number_input('Borrower Age', min_value=18, max_value=100)

education = st.selectbox('Education Level', ['High School or Below', 'college', 'Bechalar',
'Master or Above'])

gender = st.selectbox('Gender', ['male', 'female'])

if st.button('Predict Loan Status'):

prediction = model.predict(input_data)[0]

st.write('### Loan Status:', prediction)

from pyngrok import ngrok

import os

ngrok.kill()

from pyngrok import ngrok

ngrok.set_auth_token("2xOvwc6Ll7lb3PTEntOvfmP8tXu_7yqmeQzPw3FroK2UEoUL9")

public_url = ngrok.connect("http://localhost:8501")

print(f"Public URL: {public_url}")

!streamlit run app.py &> /dev/null &

print("Streamlit app is live at: https://23af-35-236-207-5.ngrok-free.app")

from pyngrok import ngrokfrom pyngrok import ngrok

ngrok.set_auth_token("2xOvwc6Ll7lb3PTEntOvfmP8tXu_7yqmeQzPw3FroK2UEoUL9")

```

```
public_url = ngrok.connect("http://localhost:8501")
```

```
print(f"Streamlit app is live at: {public_url}")
```

```
Streamlit app is live at: NgrokTunnel: "https://7a18-34-73-241-76.ngrok-free.app" ->  
"http://localhost:8501"
```